MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

LOGIC CONVERTER, DIGITAL TO ANALOG, KEY CODE DBC

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 SCOPE. This document establishes the requirements and test procedures for the acquisition of an electronic digital to analog logic converter module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Naval Electronic Systems Command, ATTN. ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 6.377 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirement specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DBC.

3.5.2 Size. The module size shall be 1C.

3.5.3 Weight. The module shall have a maximum weight of 105 grams.

3.5.4 Contact pins. The module shall have 60 contact pins (60 optional) positioned A1 through A20 and C1 through C40 (A21 through A40 and B1 through B40 optional).

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3 5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/155-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for power supply voltages 5.0 Vdc ±5 percent, +15 Vdc ±5 percent, and -15 Vdc ±3 percent.

4. QUALITY ASSURANCE PROVISIONS

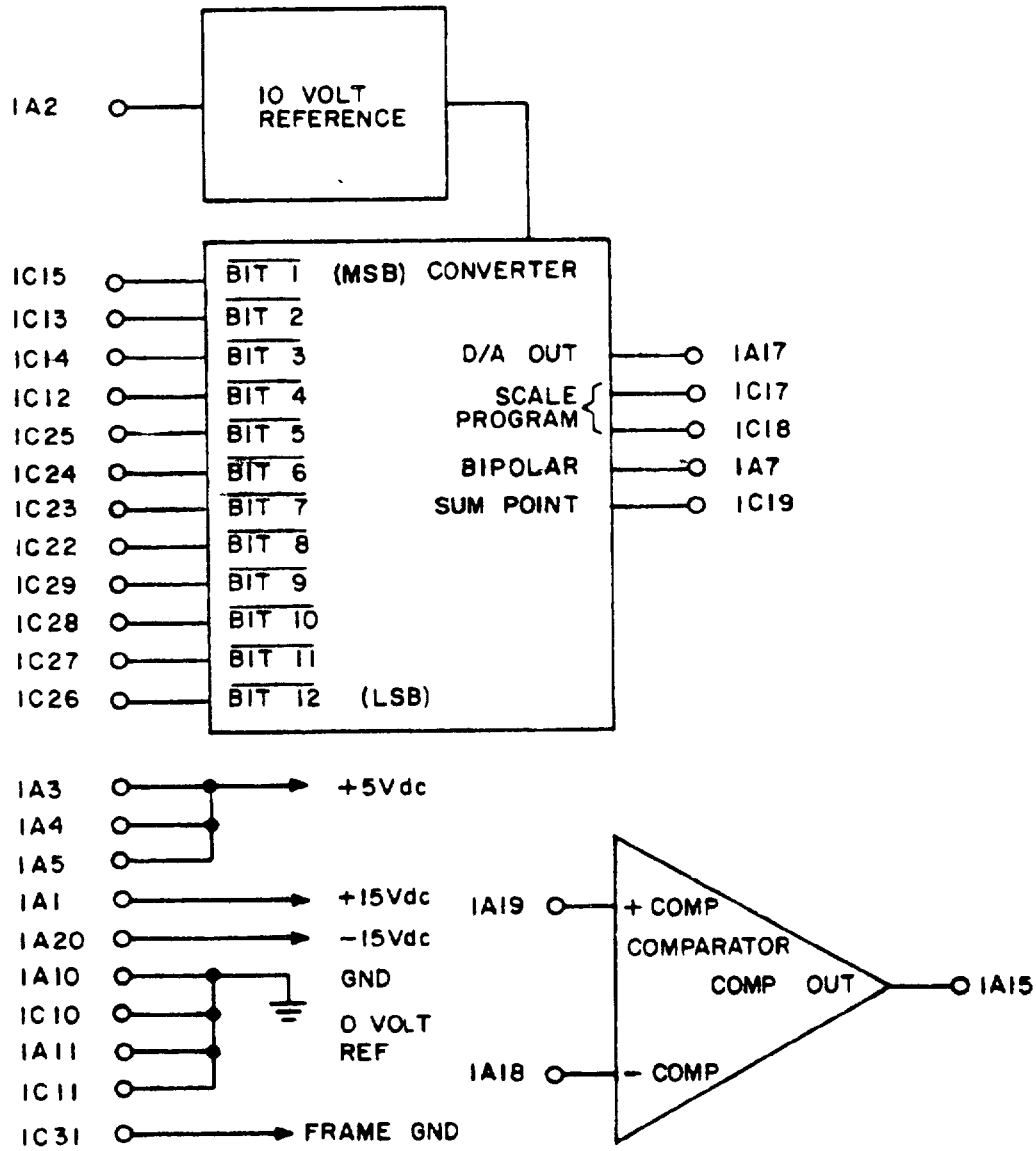
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTES:
1. BIT = True bit input
2. $\overline{\text{BIT}}$ = Inverted bit input.
3. D/A OUT = Digital to analog output.
4. SCALE PROGRAM = Scale programming pin.
5. BIPOLAR = Bipolar programming pin.
6. SUM POINT = Summing point.
7. +COMP = Positive comparator input.
8. -COMP = Negative comparator input.
9. COMP OUT = Comparator output.
10. MSB = Most significant bit.
11. LSB = Least significant bit.
12. Pins 1A6, 1A8, 1A9, 1A12-1A14, 1A16, 1C1-1C9, 1C16, 1C20, 1C21, 1C30, and 1C32-1C40 are unused. Pins 1A21-1A40 and 1B1-1B40 are optional

FIGURE 1. DBC module function diagram, digital to analog logic converter.

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* TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	Stress	III	S	9 7	10 3	9 7	10 3	9 7	10 3	9 7	10 3	V
A2	Stress	III	S	9 9	10 1	9 9	10 1	9 9	10 1	9.9	10 1	V
A3	Stress	III	S	0 01	0 8	0 01	0 8	0 01	0 8	0 01	0 8	V
B1-B12	Input breakdown current	IV	S	0 0	1 0	0 0	1 0	0 0	1.0	0.0	1 0	mA
C1,C2	Isolation	V	S	10	∞	1 0	∞	10	∞	1.0	∞	MΩ
D1-D12	Input current, low level	VI	X	0 1	1.73	0 1	1.73	0 1	1.73	0.1	1 73	mA
E1-E12	Input current, high level	VI	X	-100	100	-100	100	-100	100	-100	100	µA
F1,F2	10 volt reference	VII	X	9 9975	10 0025	9 9950	10 0050	9 9950	10 0050	9 9950	10 0050	V
G1	Output voltage (D/A, 0-10 V scale, binary no 0)	VIII	X	9 9938	10 0013	9 9913	10 0038	9 9913	10 0038	9 9913	10 0038	V
G2	Output voltage (D/A, 0-10 V scale, binary no. 4095)	VIII	X	-0 0012	0 0012	-0 0012	0 0012	-0 0012	0 0012	-0 0012	0 0012	V
H1	Output voltage (D/A, 0-5 V scale, binary no. 0)	VIII	X	4.9963	5 0013	4 9951	5.0025	4 9951	5 0025	4 9951	5 0025	V
H2	Output voltage (D/A, 0-5 V scale, binary no. 4095)	VIII	X	-0 0012	0 0012	-0.0012	0 0012	-0 0012	0 0012	-0 0012	0 0012	V
J1	Output voltage (D/A, -10 to +10 V scale, binary no. 0)	VIII	X	9 9902	10 0001	9 9877	10 0026	9 9877	10.0026	9.9877	10 0026	V
J2	Output voltage (D/A, -10 to +10 V scale, binary no 4095)	VIII	X	-10 0050	-9 9951	-10 0074	-9 9926	-10 0074	-9 9926	-10 0074	-9 9926	V
K1	Output voltage (D/A, -5 to +5 V scale, binary no 0)	VIII	X	4 9951	5 0000	4.9938	5.0013	4.9938	5.0013	4 9938	5 0013	V
K2	Output voltage (D/A, -5 to +5 V scale, binary no 4095)	VIII	X	-5 0025	-4 9975	-5 0037	-4.9963	-5 0037	-4 9963	-5 0037	-4 9963	V
L1	Power supply current, low level (+15 V)	IX	X	1	45	1	45	1	45	1	45	mA
L2	Power supply current, low level (+5 V)	IX	X	1	60	1	60	1	60	1	60	mA

See footnotes at end of table

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* TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L3	Power supply current, low level (-15 V)	IX	X	1	60	1	60	1	60	1	60	mA
L4	Power supply current, high level (+15 V)	IX	X	1	45	1	45	1	45	1	45	mA
L5	Power supply current, high level (+5 V)	IX	X	1	60	1	60	1	60	1	60	mA
L6	Power supply current, high level (-15 V)	IX	X	1	45	1	45	1	45	1	45	mA
M1	Comparator output voltage, high level	X	X	2.4	4 75	2 4	4 75	2.4	4 75	2 4	4 75	V
M2	Comparator output voltage, low level	X	X	0.0	0 4	0 0	0 4	0.0	0 4	0 0	0 4	V
N1-N12	Settling time check (0-10 V)	XI	S	← See table XI →								
P	Settling time check (0-5 V)	XI	S	← See table XI →								
Q	Settling time check (-10 to +10 V)	XI	S	← See table XI →								
R	Settling time check (-5 to +5 V)	XI	S	← See table XI →								
S1-S4	Comparator response time	XII	S	50	400	50	400	50	400	50	400	ns
T1,T2	Power supply filter capacitance (+15 V and -15 V)	XIII	S	1.6	3 0	1 6	3 0	1.6	3 0	1 6	3 0	µF
T3	Power supply filter capacitance (+5 V)	XIII	S	3 2	6 2	3.2	6.2	3.2	6.2	3 2	6 2	µF
U1-U3	Power supply transient amplitude (+15 V, -15 V, and +5 V)	XIV	Q	0	200	0	200	0	200	0	200	mV
U4	Power supply transient amplitude (10 V REF)	XIV	Q	0	100	0	100	0	100	0	100	mV
V1-V14	Input capacitance	XV	Q	2	15	2	15	2	15	2	15	pF
W1-W6	Continuity	XVI	S	0	1	0	1	0	1	0	1	Ω

See footnotes at end of table.

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* TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X1-X4	D/A output stability check	XVII	S	← No oscillation →								
Y	Absolute accuracy	XVIII	X	-2	+2	-2	+2	-2	+2	-2	+2	LSB
Z	Monotonicity	XVIII	X	-3/4	+1/2	-3/4	+1/2	-3/4	+1/2	-3/4	+1/2	LSB
AA	Linearity	XVIII	X	-1	+1	-1	+1	-1	+1	-1	+1	LSB

1/ "X" parameters shall be 100 percent tested during module acceptance tests
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirement of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows A, B, C where applicable, followed by the remaining tests in the sequence. If test conditions A and B are successively repeated, the time interval between execution of each of these test conditions should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature test. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5 1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5 3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following life test and while still at the maximum class temperature, electrical tests D through U and W through AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through AA shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331 and as follows:

a. Absolute accuracy. Absolute accuracy is the maximum allowable deviation between the actual and ideal analog output for a given digital inut.
b. Monotonicity. The analog output of a monotonic converter will, in response to a continuously increasing input count, increase within defined limits.
c. Linearity. The nonlilnearity of a converter is the deviation from the best straight line approximation of the analog response over the entire range of the digital input.
d. LSB. One LSB is the weight of the least significant digital input bit and the analog output voltage increment to which it ideally corresponds.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

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* 6.3 Cross reference of classification. Part number M28787/155-1 supersedes part number M28787/155. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	
D/A output at 10 V	1.65 W (outputs not loaded)
D/A output at 0 V	1.875 W (outputs not loaded)
Supply voltages	+15 Vdc ±5 percent -15 Vdc ±3 percent +5 Vdc ±5 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current	
Low level (V_{IL} = 0 V)	1.73 mA maximum
High level (V_{IH} = 2.4 V)	100 µA maximum
High level (V_{IH} = 5.25 V)	1.0 mA maximum
Output voltage scales	0 to +10 V 0 to +5 V -10 to +10 V -5 to +5 V
Output drive current	
D/A output	5 mA maximum
Reference output	3 mA maximum
Comparator output	
High level (V_{OH} = 2.4 V)	800 µA maximum
Low level (V_{OL} = 0.4 V)	6 mA maximum
Comparator output voltage	
Low level (I_{OL} = 6 mA)	0.4 V maximum
High level (I_{OH} = 800 µA	2.4 V minimum
Settling time (to 1/2 LSB accuracy of 12 bits, 0.012 percent)	
Most significant bit only	2.5 µs maximum
All 12 bits driven	3.5 µs maximum
Comparator response time	400 ns maximum
Reference voltage	10.0 Vdc ±0.005 percent
Output short circuit current	D/A output is not short circuit protected.
Noise spike suppression	Module does not have noise spike suppression

* 6.5 Module function. This module contains a 12-bit converter, a 10 volt reference source, and an independent analog comparator. The output voltage of the converter is controlled by the 12-digital inputs. The combinations of low-and-high levels on the inputs cause corresponding voltage level changes in the analog output. The analog output range is pin programmable according to table II. The 10-volt precision reference source provides the high accuracy needed in the digital to analog conversion and for bipolar operation. The independent analog comparator, although not used for digital to analog conversions, is available for use in conjunction with the DBA module to perform analog to digital functions. The inputs to the analog comparator should be connected to ground when not used.

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6.6 Input to output relationship. The digital inputs are weighted as a 12-bit binary number is weighted. Thus, 4096 combinations of zeros and ones can be applied to the inputs, causing 4096 different output voltage levels on the analog output. When a low level is applied to an input, an increase in voltage is caused on the analog output. The analog output increases in 4096 increments from the minimum to the maximum of the scale being used. The increment has the approximate value of the least significant bit (LSB). The value of the LSB for each voltage scale is listedin table II. For the converter, when connected for bipolar operation, the analog output will be approximately zero volt when the digital input is decimal 2048 (equivalent to a digital input word 011111111111). This will enable the most significant bit (MSB) to be used as the sign bit (i.e., if the MSB is "1", then the analog output will be negative, if the MSB is "0", then the output will be positive).

*TABLE II. Output relationships. 1/, 3/

Output scale	Pin program (connect pin to pin)	Analog output			
		Resolution (LSB)	Minimum	Maximum	Tolerance (±1/2 LSB)
0 to 10 V	1A17 to 1C18	0.00244 V	0.00000 V	9 99756 V	±0.00122 V
0 to 5 V	1A17 to 1C18 1C17 to 1C19	0.00122 V	0.00000 V	4.99878 V	±0.00122 V 2/
-10 to +10 V	1A17 to 1C17 1A7 to 1C19	0.00488 V	-10.00000 V	9.99512 V	±0.00244 V
-5 to +5 V	1A7 to 1C19 1A17 to 1C18	0.00244 V	-5.0000 V	4.99756 V	±0.00122 V

1/ Resolution = LSB = Full scale voltage - 4096.
2/ Tolerance = Relative accuracy = ±1/2 LSB (for 0 to 5 V operation relative accuracy = ±1 LSB).
3/ Maximum output = Full scale -LSB.

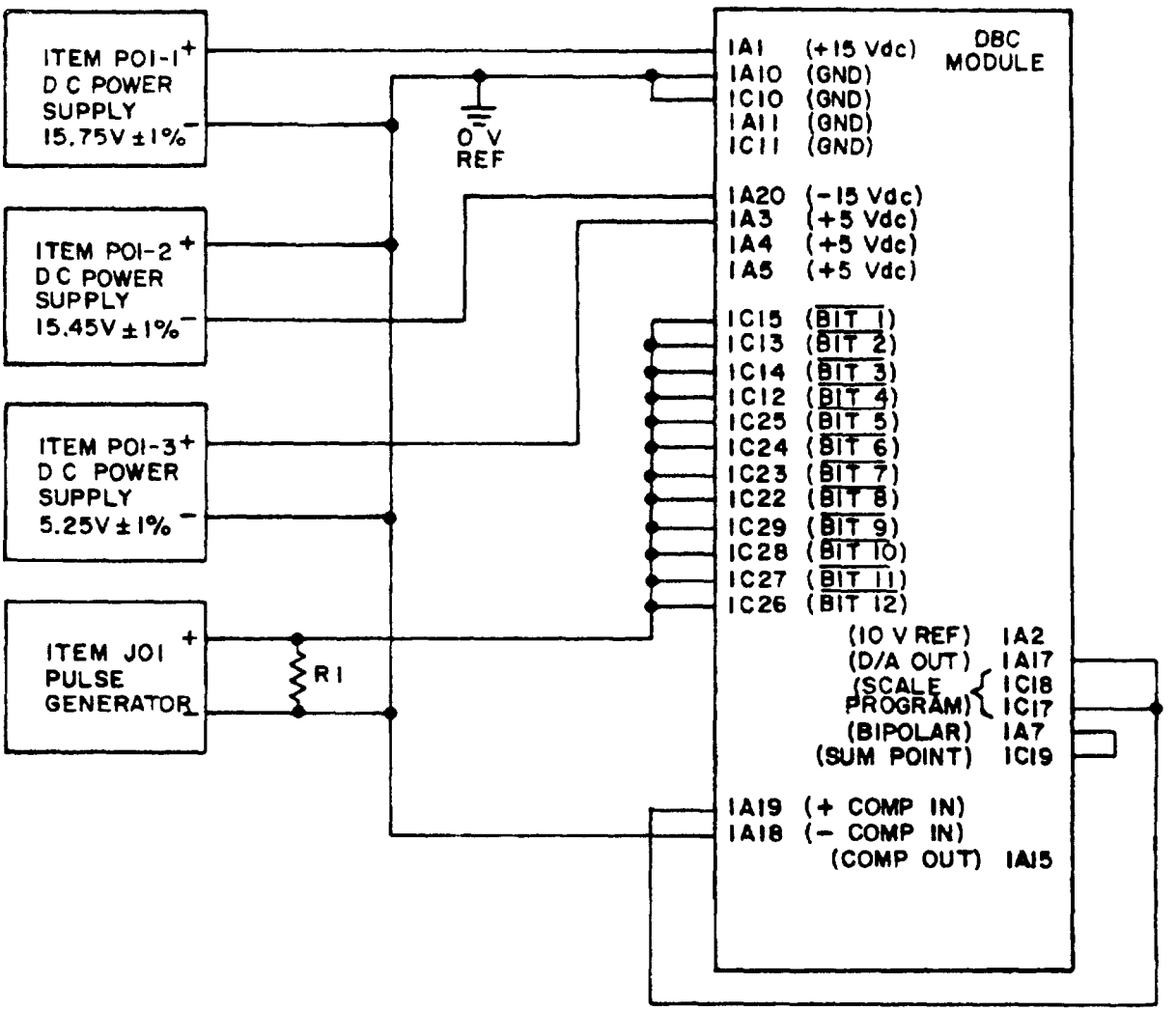
6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document baaed on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES
1. R1 = 51.1Ω ±5%, 1 W.
2. Pulse generator setup

   a. PW = 4 ±0.4 µs.
   b. PRP = 8 ±0.8 µs.
   c. $t_{TLH} = t_{THL}$ = 25 ±25 ns.
   d. Amplitude = 4.0 ±0.5 V

FIGURE 2 Operating temperature and life test circuit.

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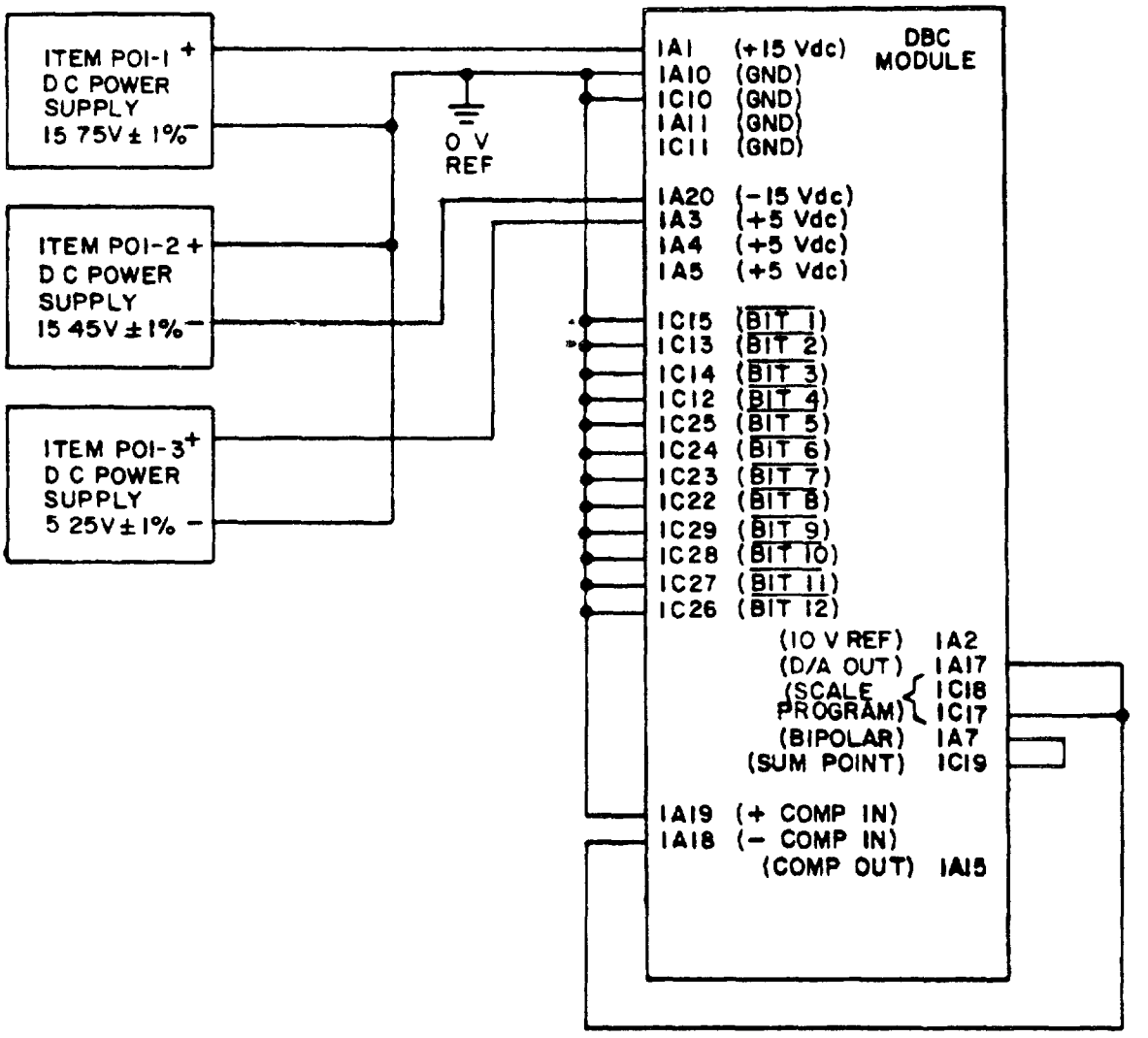


FIGURE 3. Component temperature test circuit.

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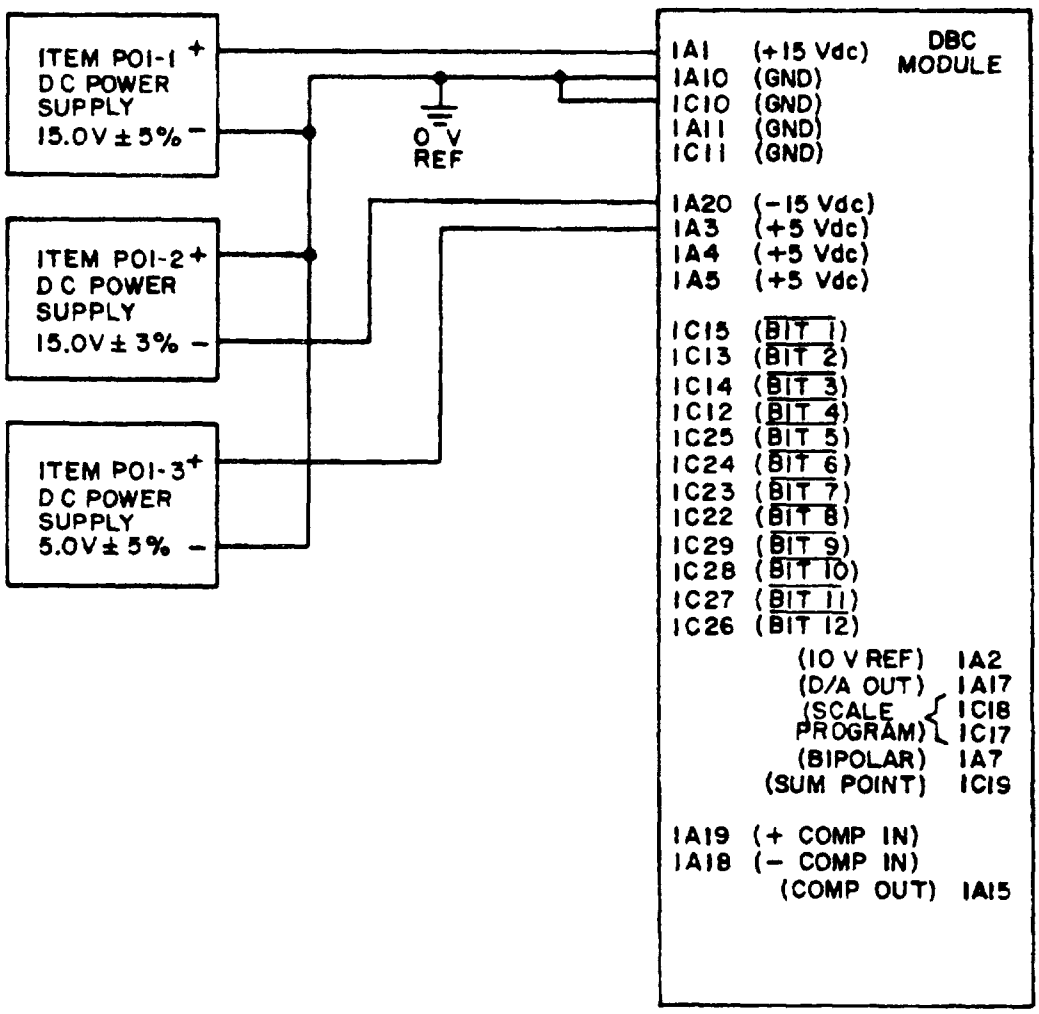


FIGURE 4. Vibration test circuit.

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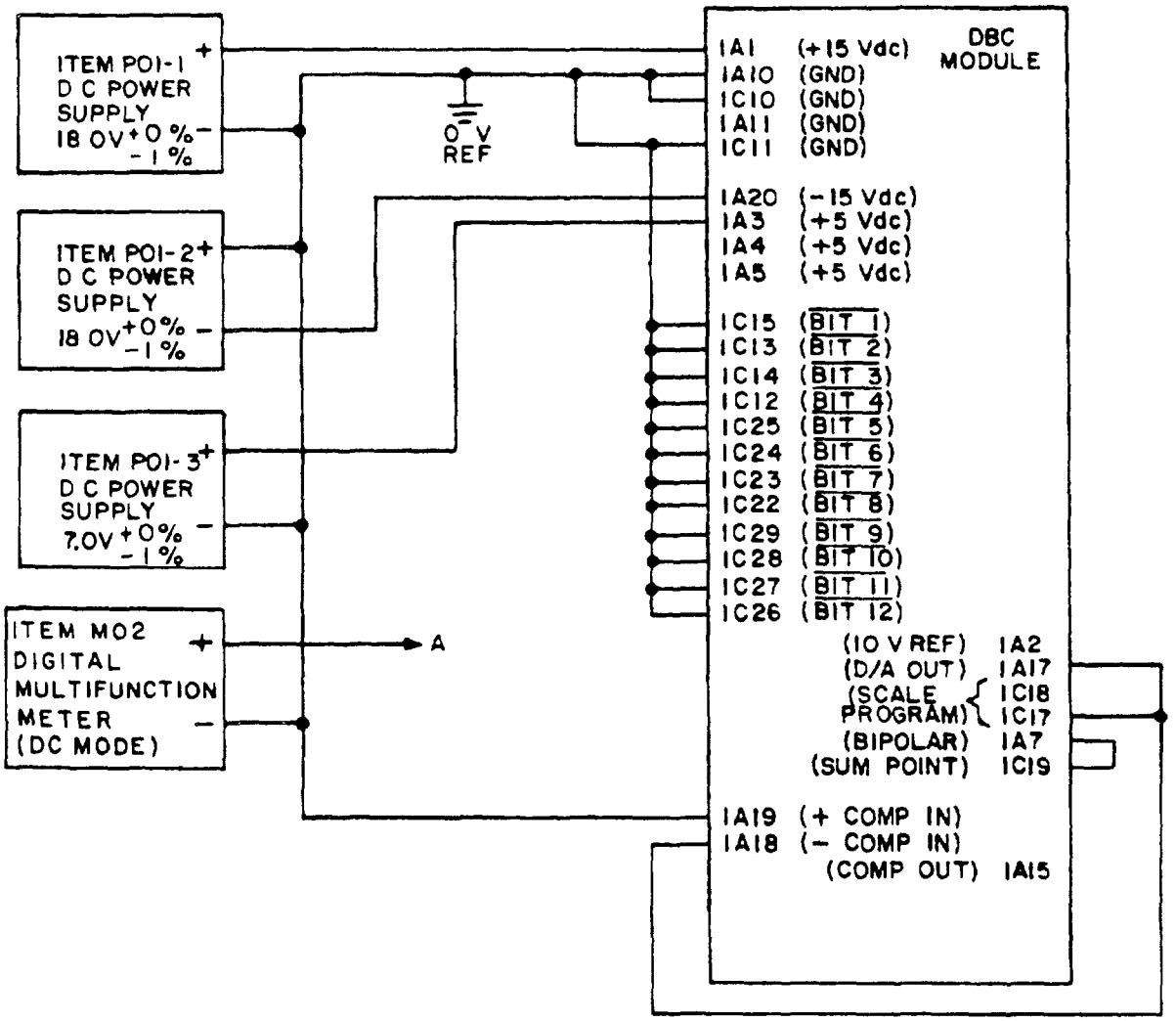
TABLE III. Stress test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	D/A OUT Stress	1A17	9.7	10.3	V	5
A2	10 V REF Stress	1A2	9.9	10.1	V	5
A3	COMP OUT Stress	1A15	0.01	0.8	V	5

TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
B1	$\overline{\text{BIT 1}}$ Input breakdown current	1C15	0.0	1.0	mA	6
B2	$\overline{\text{BIT 2}}$	1C13	↓	↓	↓	↓
B3	$\overline{\text{BIT 3}}$	1C14				
B4	$\overline{\text{BIT 4}}$	1C12				
B5	$\overline{\text{BIT 5}}$	1C25				
B6	$\overline{\text{BIT 6}}$	1C24				
B7	$\overline{\text{BIT 7}}$	1C23				
B8	$\overline{\text{BIT 8}}$	1C22				
B9	$\overline{\text{BIT 9}}$	1C29				
B10	$\overline{\text{BIT 10}}$	1C28				
B11	$\overline{\text{BIT 11}}$	1C27				
B12	$\overline{\text{BIT 12}}$	1C26				

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NOTES:
1. For test condition A2, load the reference (pin 1A2) with a 3 32 kΩ ±1%, 1/8 W, resistor connected to GND
2. Measurements shall be recorded after point A has been connected for at least 10 seconds

FIGURE 5 Stress test circuit for table III.

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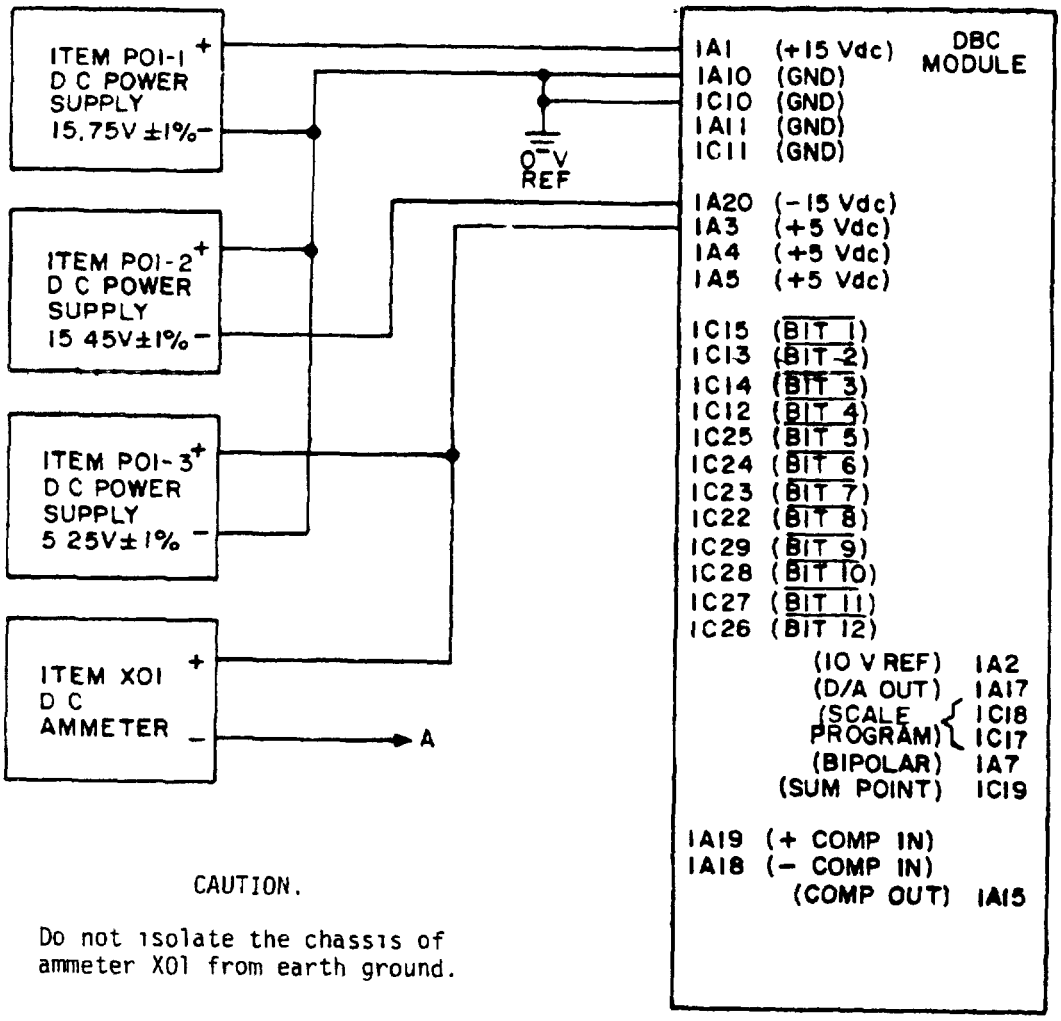


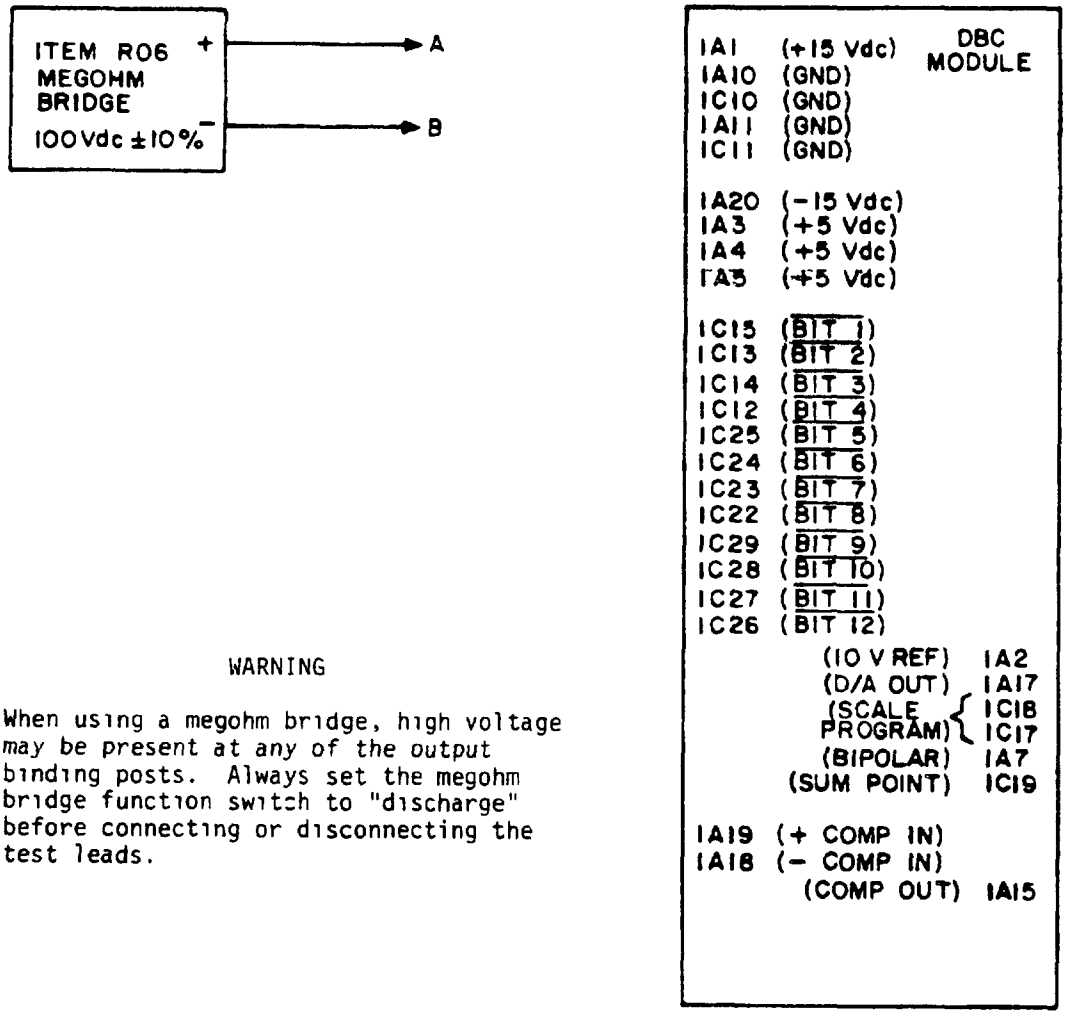
FIGURE 6. Input breakdown current test circuit for table IV

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TABLE V Isolation test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, used to unused pins	1A1,1A2,1A3,1A4,1A5, 1A7,1A10,1A11,1A15, 1A17,1A18,1A19,1A20, 1C10,1C11,1C12,1C13, 1C14,1C15,1C17,1C18, 1C19,1C22,1C23,1C24, 1C25,1C26,1C27,1C28, 1C29,1C31	1A6,1A8,1A9,1A12, 1A13,1A14,1A16,1A21, 1A22,1A23,1A24,1A25, 1A26,1A27,1A28,1A29, 1A30,1A31,1A32,1A33, 1A34,1A35,1A36,1A37, 1A38,1A39,1A40,1C1, 1C2,1C3,1C4, 1C5, 1C6, 1C7,1C8,1C9,1C16, 1C20,1C21,1C30,1C32, 1C33,1C34,1C35,1C36, 1C37,1C38,1C39,1C40	10	∞	MΩ	7
C2	Isolation, all pins to frame	All pins except 1C31	1C31	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

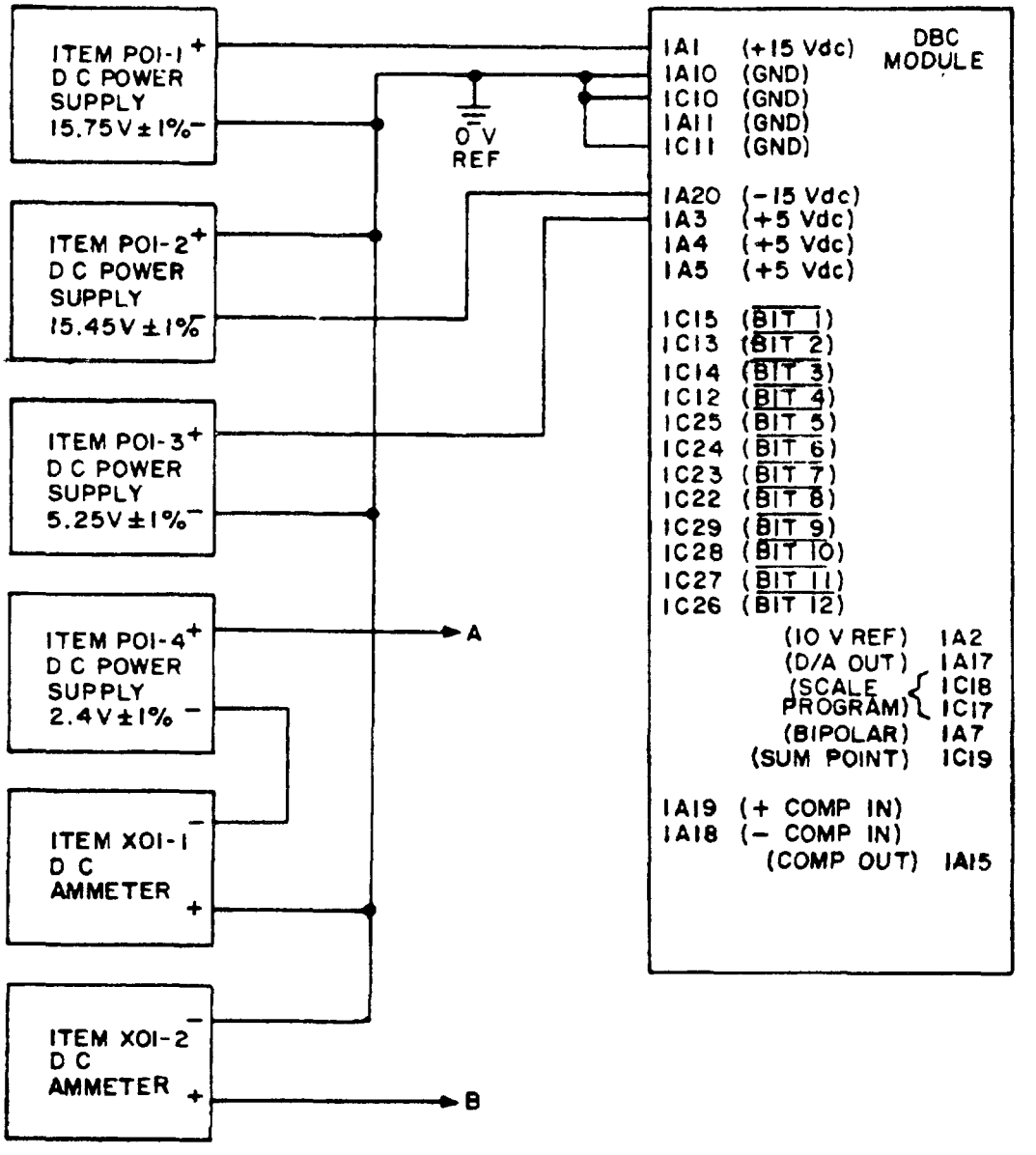
FIGURE 7. Isolation test circuit for table V

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TABLE VI Input current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	$\overline{\text{BIT 1}}$ Input current, low level	Open	1C15	0.1	1.73	mA	8
D2	$\overline{\text{BIT 2}}$		1C13				
D3	$\overline{\text{BIT 3}}$		1C14				
D4	$\overline{\text{BIT 4}}$		1C12				
D5	$\overline{\text{BIT 5}}$		1C25				
D6	$\overline{\text{BIT 6}}$		1C24				
D7	$\overline{\text{BIT 7}}$		1C23				
D8	$\overline{\text{BIT 8}}$		1C22				
D9	$\overline{\text{BIT 9}}$		1C29				
D10	$\overline{\text{BIT 10}}$		1C28				
D11	$\overline{\text{BIT 11}}$		1C27				
D12	$\overline{\text{BIT 12}}$		1C26				
E1	$\overline{\text{BIT 1}}$ Input current, high level	1C15	Open	-100	100	µA	
E2	$\overline{\text{BIT 2}}$	1C13					
E3	$\overline{\text{BIT 3}}$	1C14					
E4	$\overline{\text{BIT 4}}$	1C12					
E5	$\overline{\text{BIT 5}}$	1C25					
E6	$\overline{\text{BIT 6}}$	1C24					
E7	$\overline{\text{BIT 7}}$	1C23					
E8	$\overline{\text{BIT 8}}$	1C22					
E9	$\overline{\text{BIT 9}}$	1C29					
E10	$\overline{\text{BIT 10}}$	1C28					
E11	$\overline{\text{BIT 11}}$	1C27					
E12	$\overline{\text{BIT 12}}$	1C26					

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CAUTION Do not isolate the chassis of ammeter X01 from earth ground.

NOTE Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

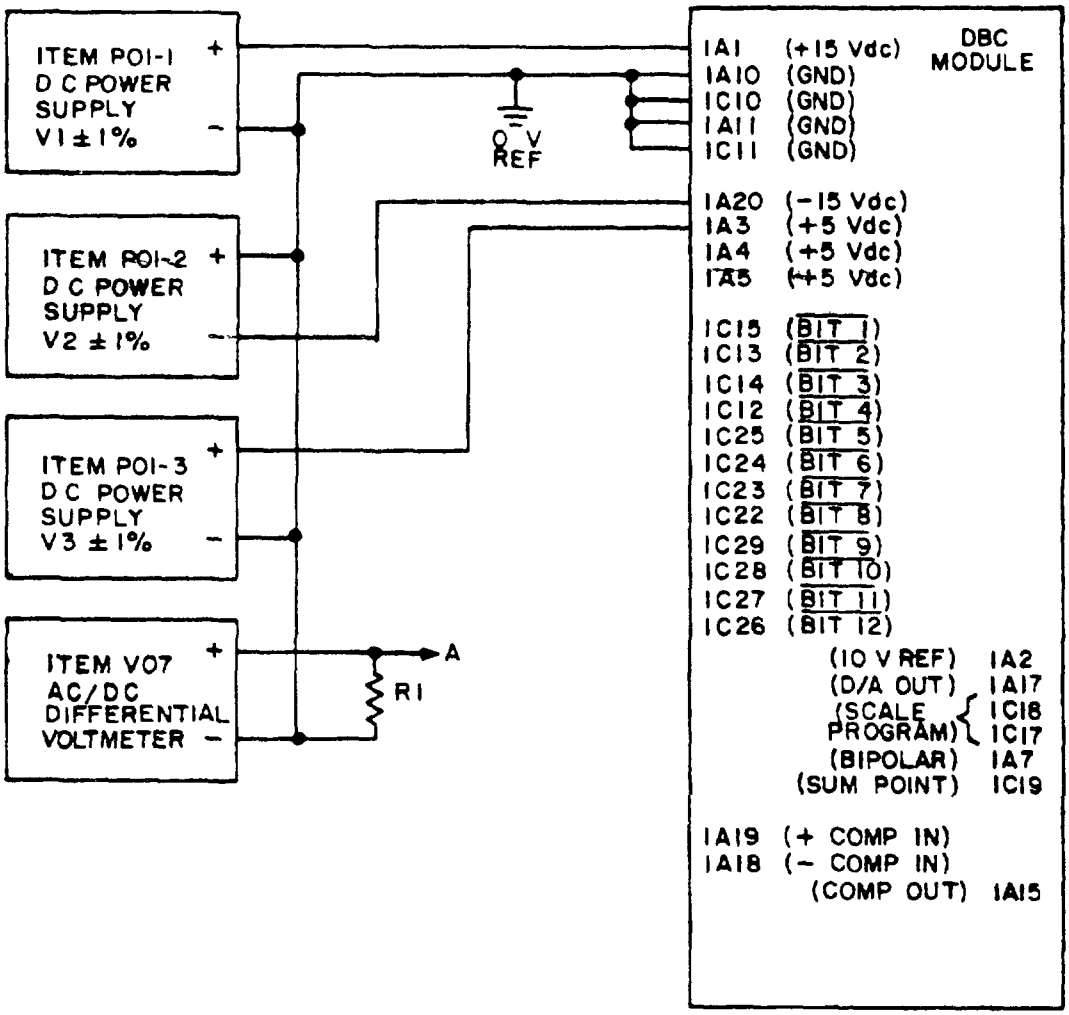
FIGURE 8. Input current (low-level and high-level) test circuit for table VI

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* TABLE VII. 10 volt reference test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		V1 to	V2 to	V3 to	A to pin	Min	Max		
F1	10 volt reference, power supplies maximum	15.75 V	15.45 V	5.25 V	1A2	9.9975	10.0025	V	9
F2	10 volt reference, power supplies minimum	14.25 V	14 55 V	4.75 V	1A2	9.9975	10.0025	V	9

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NOTES
1. Adjust the voltage supplies V1, V2, and V3 according to table VII for each test condition
2 R1 = 3.32 kΩ ±1%, 1/8 W
3. Allow a minimum of 5 minutes warmup time before making measurements

FIGURE 9. 10 volt reference test circuit for table VII

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* TABLE VIII Output voltage (input threshold conditions) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin to pin	Min	Max		
G1	Output voltage (D/A, 0-10 V scale, binary no. 0)	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	Open	1C18 to 1A17	9.9938	10.0013	V	10
G2	Output voltage (D/A, 0-10 V scale, binary no. 4095)	Open	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	1C18 to 1A17	-0.0012	0.0012		
H1	Output voltage (D/A, 0-5 V scale, binary no. 0)	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	Open	1C18 to 1A17, 1C19 to 1C17	4 9963	5.0013		
H2	Output voltage (D/A, 0-5 V scale, binary no. 4095)	Open	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	1C18 to 1A17, 1C19 to 1C17	-0 0012	0 0012		
J1	Output voltage (D/A, -10 to +10 V scale, binary no. 0)	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	Open	1A7 to 1C19, 1C17 to 1A17	9 9902	10.0001		
J2	Output voltage (D/A, -10 to +10 V scale, binary no. 4095)	Open	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	1A7 to 1C19, 1C17 to 1A17	-10.0050	-9 9951		

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* TABLE VIII Output voltage (input threshold conditions) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin to pin	Min	Max		
K1	Output voltage (D/A, -5 to +5 V scale, binary no. 0)	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	Open	1A7 to 1C19, 1C18 to 1A17	4.9951	5.0000	V	10
K2	Output voltage (D/A, -5 to +5 V scale, binary no. 4095)	Open	1C15,1C13, 1C14,1C12, 1C25,1C24, 1C23,1C22, 1C29,1C28, 1C27,1C26	1A7 to 1C19, 1C18 to 1A17	-5.0025	-4.9975	V	10

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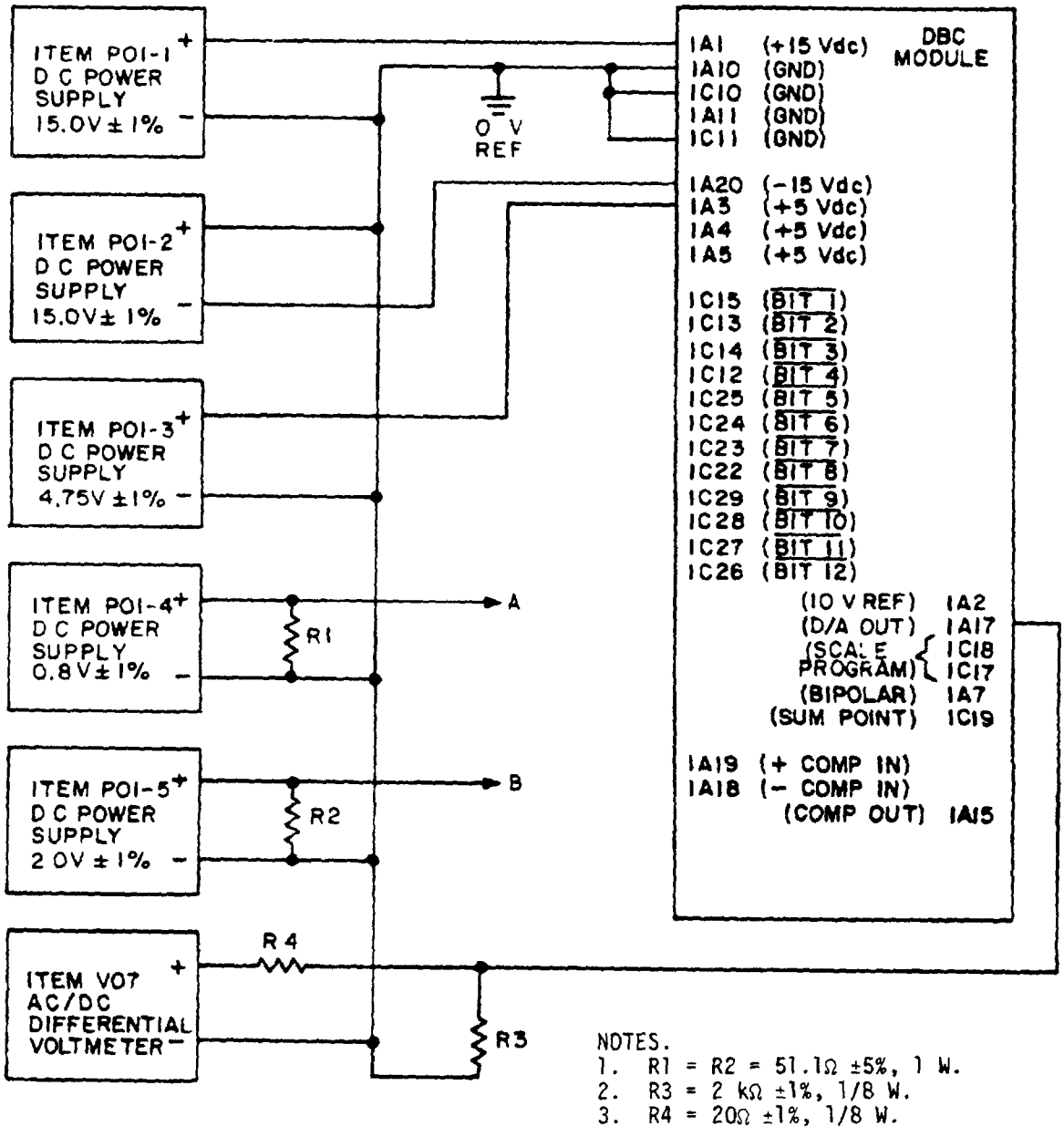


FIGURE 10 Output voltage (input threshold conditions) test circuit for table VIII.

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TABLE IX. Power supply current (low-level and high-level) test

Test condition	Parameter	Connect Pin 1A10 to	Read meter	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
L1	Power supply current, low level (+15 V)	1C15,1C13,1C14,1C12,1C25, 1C24,1C23,1C22,1C29,1C28, 1C27,1C26	X01-1	1	45	mA	11
L2	Power supply current, low level (+5 V)	1C15,1C13,1C14,1C12,1C25, 1C24,1C23,1C22,1C29,1C28, 1C27,1C26	X01-3		60		
L3	Power supply current, low level (-15 V)	1C15,1C13,1C14,1C12,1C25, 1C24,1C23,1C22,1C29,1C28, 1C27,1C26	X01-2		60		
L4	Power supply current, high level (+15 V)	Open	X01-1		45		
L5	Power supply current, high level (+5 V)	Open	X01-3		60		
L6	Power supply current, high level (-15 V)	Open	X01-2		45		

TABLE X Comparator output voltage (high-level and low-level) test and offset voltage check

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect Adjust Vos to	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
M1	Comparator output voltage, high level	Open	1A15	1A15	0.00125	2.4	4.75	V	12
M2	Comparator output voltage, low level	1A15	Open	1A15	-0.00125	0.0	0.4	V	12

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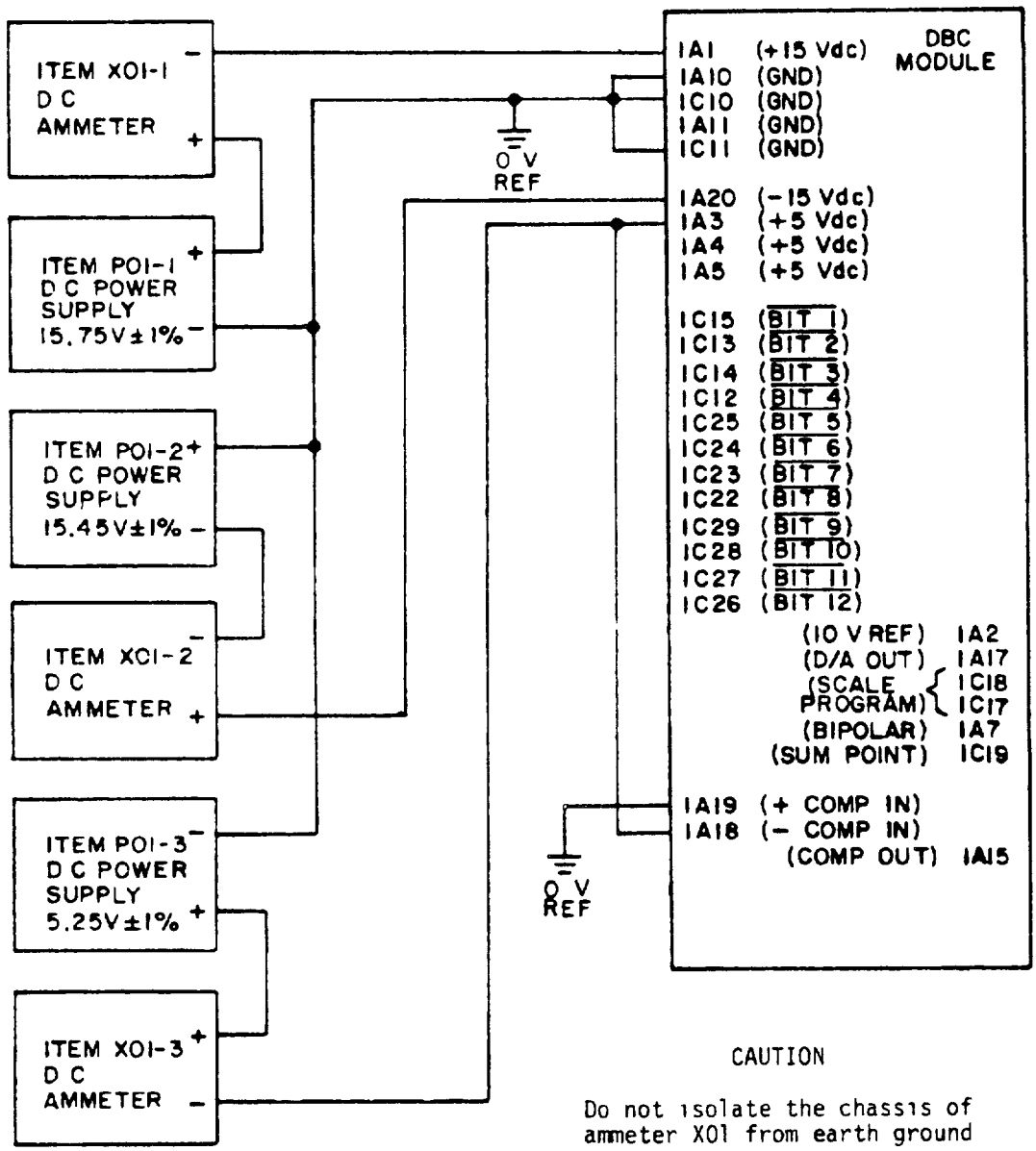
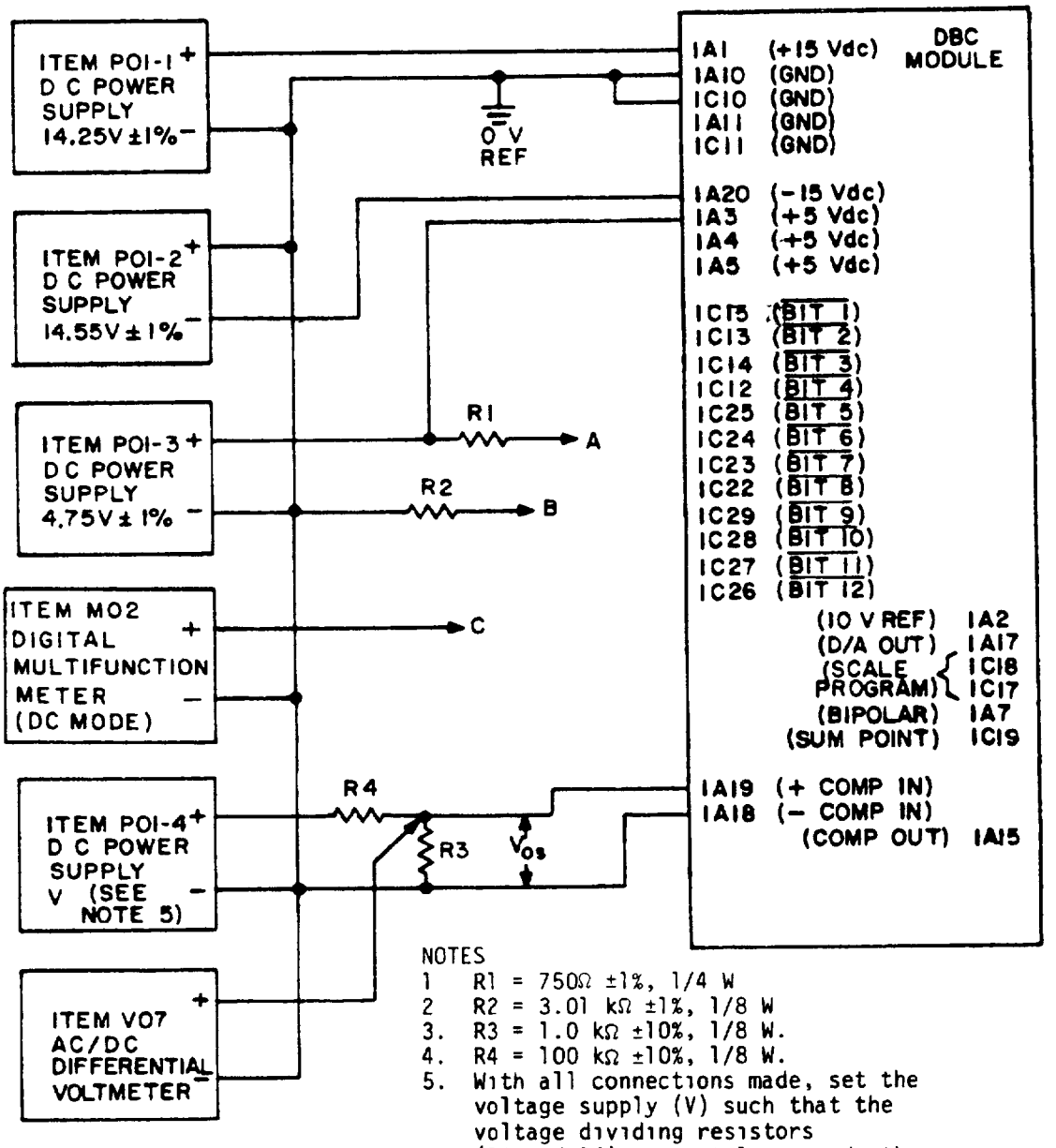


FIGURE 11 Power supply current (low-level and high-level) test circuit for table IX.

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NOTES
1 R1 = 750Ω ±1%, 1/4 W
2 R2 = 3.01 kΩ ±1%, 1/8 W
3. R3 = 1.0 kΩ ±10%, 1/8 W.
4. R4 = 100 kΩ ±10%, 1/8 W.
5. With all connections made, set the voltage supply (V) such that the voltage dividing resistors (R3 and R4) accurately provide the voltage V_{os} specified in table X at the point marked V_{os} on figure 12.
6 Adjust V_{os} to the value specified as read on item VO7.

FIGURE 12 Comparator output voltage (high-level and low-level) and offset voltage check test circuit for table X

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TABLE XI Settling time check test

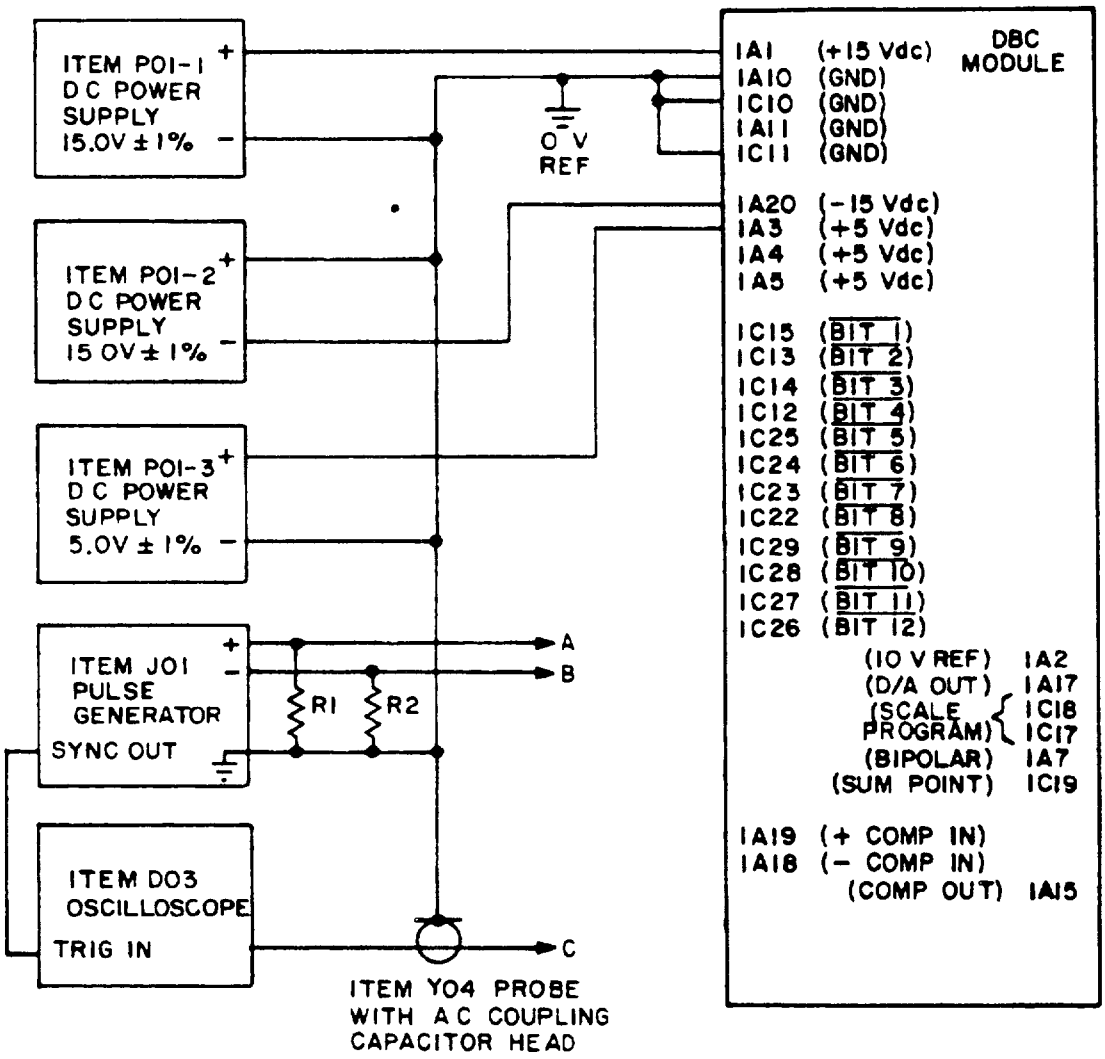
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
N1	BIT 1 Settling time check (0-10 V)	1C15	1C13, 1C14, 1C12, 1C25, 1C24, 1C23, 1C22, 1C29, 1C28, 1C27, 1C26	1A17	1C18 to 1A17	Note 4, figure 13	Note 4, figure 13	V	13, 14
N2	BIT 2	1C14	1C14, 1C12, 1C25, 1C24, 1C23, 1C22, 1C29, 1C28, 1C27, 1C26						
N3	BIT 3	1C12	1C12, 1C25, 1C24, 1C23, 1C22, 1C29, 1C28, 1C27, 1C26						
N4	BIT 4	1C11	1C25, 1C24, 1C23, 1C22, 1C29, 1C28, 1C27, 1C26						
N5	BIT 5	1C25	1C24, 1C23, 1C22, 1C29, 1C28, 1C27, 1C26						
N6	BIT 6	1C24	1C23, 1C22, 1C29, 1C28, 1C27, 1C26						
N7	BIT 7	1C23	1C22, 1C29, 1C28, 1C27, 1C26						
N8	BIT 8	1C22	1C29, 1C28, 1C27, 1C26						

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TABLE XI Settling time check test - Continued

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin	Min	Max		
N9	$\overline{\text{BIT 9}}$ Settling time check (0-10 V)	1C29	1C28,1C27, 1C26	1A17	1C18 to 1A17	Note 4, figure 13	Note 4, figure 13	V	13,14
N10	$\overline{\text{BIT 10}}$	1C28	1C27,1C26						
N11	$\overline{\text{BIT 11}}$	1C27	1C26						
N12	$\overline{\text{BIT 12}}$	1C26	Open						
P	$\overline{\text{BIT 1}}$ Settling time check (0-5 V)	1C15	1C12,1C14, 1C13,1C26, 1C22,1C23, 1C24,1C25, 1C27,1C28 1C29		1C19 to 1C17, 1C18 to 1A17				
Q	$\overline{\text{BIT 1}}$ Settling time check (-10 to +10 V)	1C15	1C12,1C14, 1C13,1C26, 1C22,1C23, 1C24,1C25, 1C27,1C28 1C29		1C19 to 1A7, 1C17 to 1A17				
R	$\overline{\text{BIT 1}}$ Settling time check (-5 to +5 V)	1C15	1C12,1C14, 1C13,1C26, 1C22,1C23, 1C24,1C25, 1C27,1C28 1C29		1C19 to 1A7, 1C18 to 1A17				

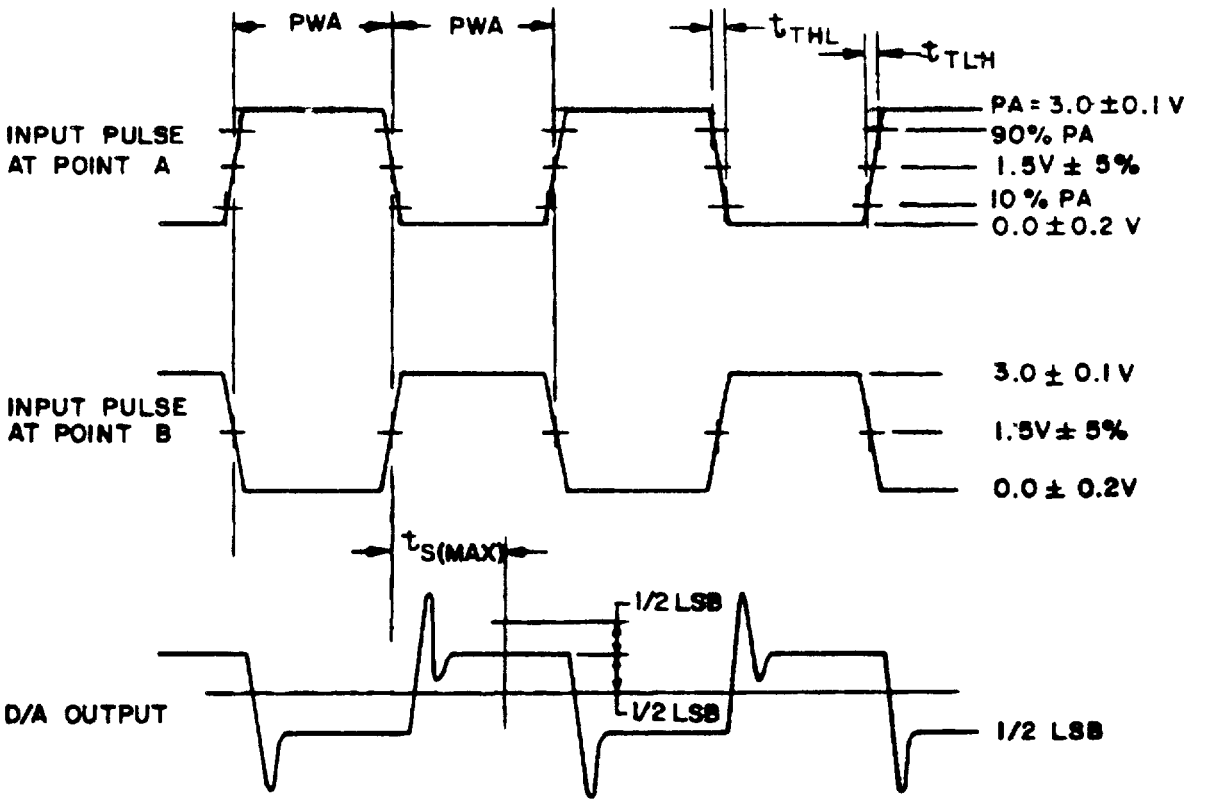
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NOTES.
1. R1 = R2 = 51.1Ω ±5%, 1 W.
2. Load the D/A output with a 2 kΩ ±1%, 1/8 W, resistor.
3 With all circuit connections made for a test condition apply the input pulses to the module The output pulse should be ac coupled into the oscilloscope so that it may be amplified for maximum amplitude or a dc voltage should be subtracted by means of a differential comparator Verify that the output pulse amplitude stays within ±1/2 LSB (±1 LSB for 0-5 V operation, test condition P) of its settled out value at 2.5 µs after the input pulse (t_S (MAX) = 2.5 µs)
The settled out value is the pulse amplitude after the ringing has stopped
4 For test conditions N1-N12, 1/2 LSB = 0.00122 V.
For test condition P, 1 LSB = 0.00122 V.
For test condition Q, 1/2 LSB = 0 00244 V.
For test condition R, 1/2 LSB = 0.00122 V.

FIGURE 13 Settling time check test circuit for table XI

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NOTE: Input pulse parameters:
PWA = 5.0 ±0.5 µs,
$t_{TLH} = t_{THL}$ = 10 ±1 ns.

* FIGURE 14 Settling time check test circuit waveforms for table XI

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TABLE XII Comparator response time test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
S1	Comparator response time (t_{PLH})	1A19	1A15	1A18	50	400	ns	15,16
S2	Comparator response time (t_{PHL})	1A19	↓	1A18	↓	↓	↓	↓
S3	Comparator response time (t_{PHL})	1A18	↓	1A19	↓	↓	↓	↓
S4	Comparator response time (t_{PLH})	1A18	↓	1A19	↓	↓	↓	↓

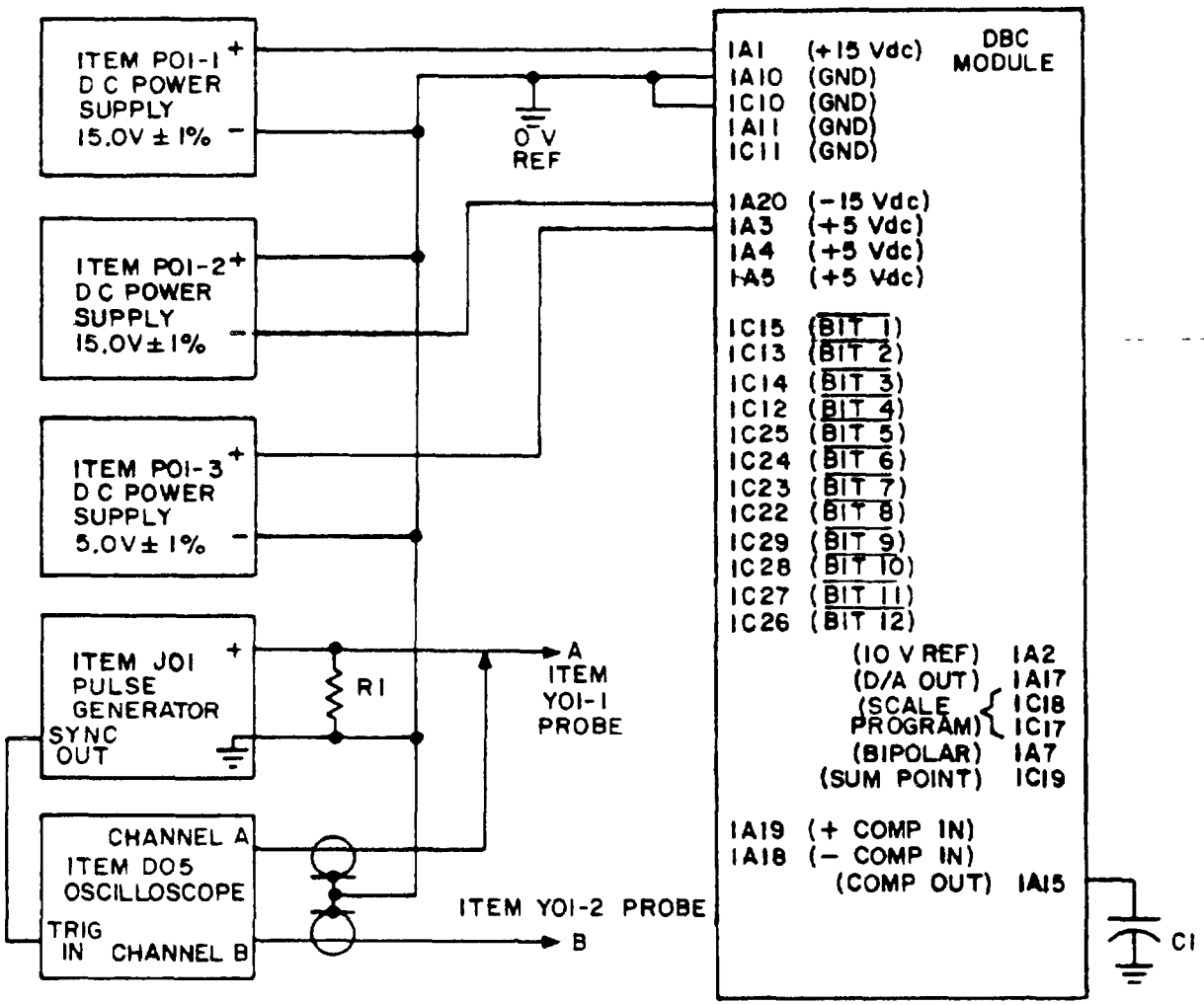
TABLE XIII Power supply filter capacitance test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Power supply filter capacitance (+15 V)	1A1	1A10,1C10	1.6	3.0	µF	17
T2	Power supply filter capacitance (-15 V)	1A10,1C10	1A20	1.6	3.0	µF	17
T3	Power supply filter capacitance (+5 V)	1A3,1A4,1A5	1A10,1C10	3.2	6.2	µF	17

TABLE XIV Power supply transient amplitude test

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
U1	Power supply transient amplitude (+15 V)	1A1	0	200	mV	18,19
U2	Power supply transient amplitude (-15 V)	1A20	↓	200	↓	↓
U3	Power supply transient amplitude (+5 V)	1A3	↓	200	↓	↓
U4	Power supply transient amplitude (10 V REF)	1A2	↓	100	↓	↓

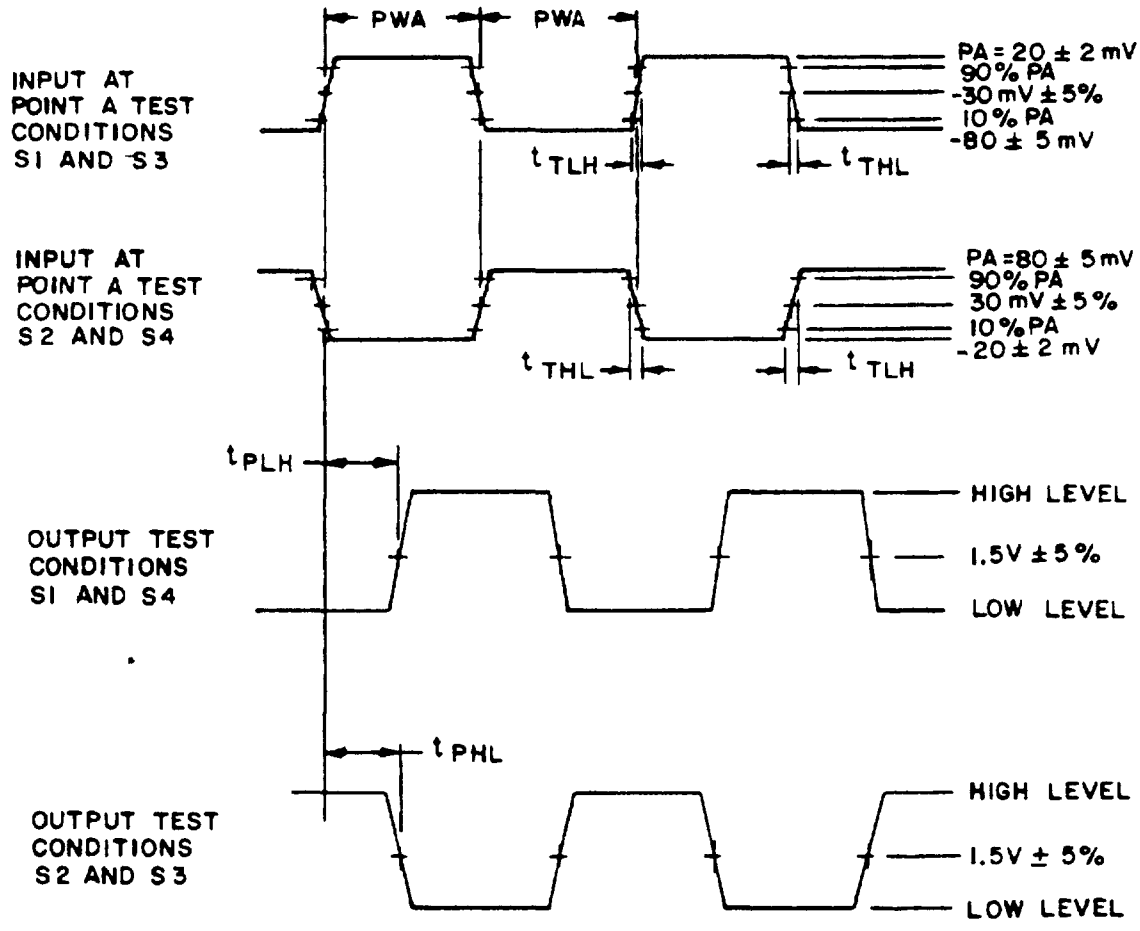
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NOTES
1 R1 = 51 1Ω ±5%, 1 W
2 C1 = 30 pF ±5%

FIGURE 15 Comparator response time test circuit for table XII

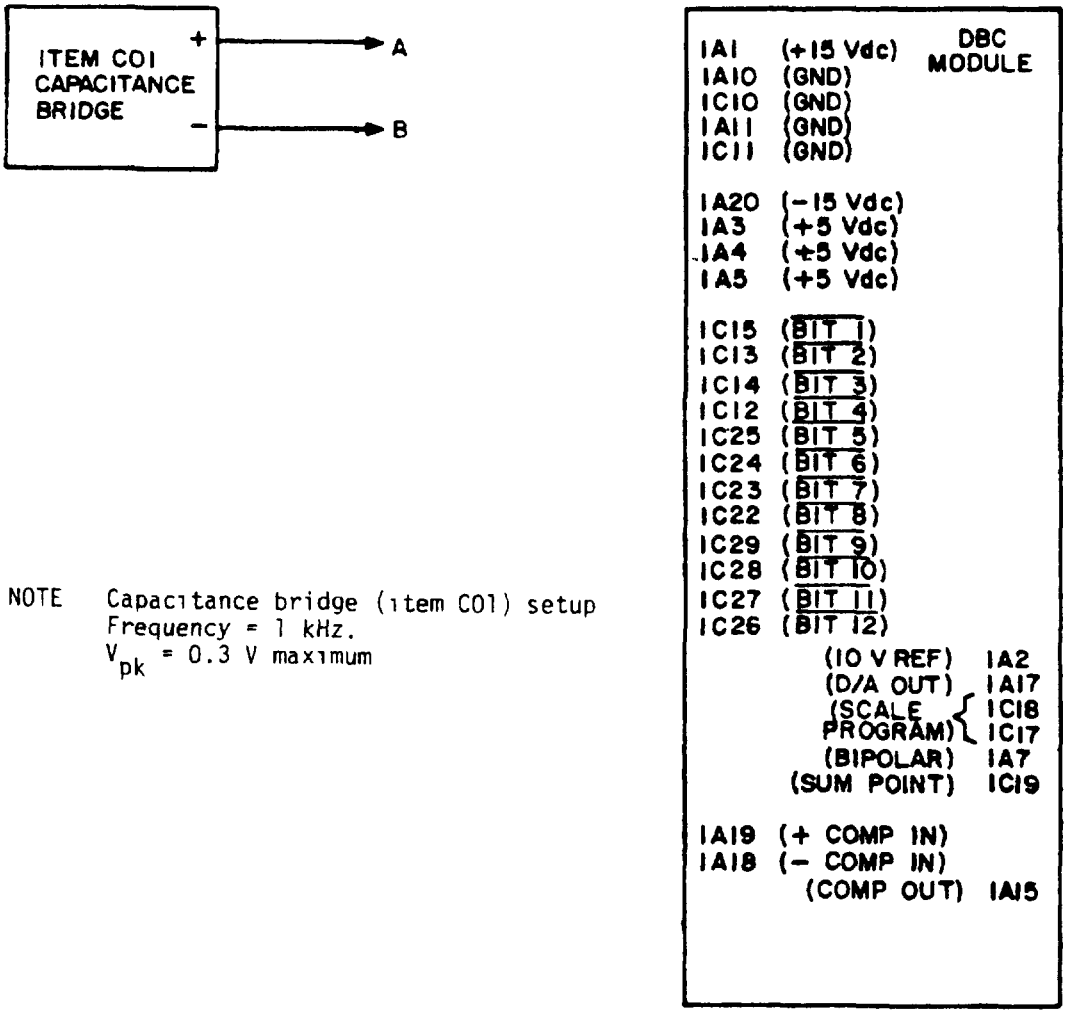
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NOTE Input pulse parameters
PWA = 1 ±0.1 µs.
$t_{TLH} = t_{THL}$ = 20 ±4 ns.

FIGURE 16. Comparator response time test circuit waveforms for table XII.

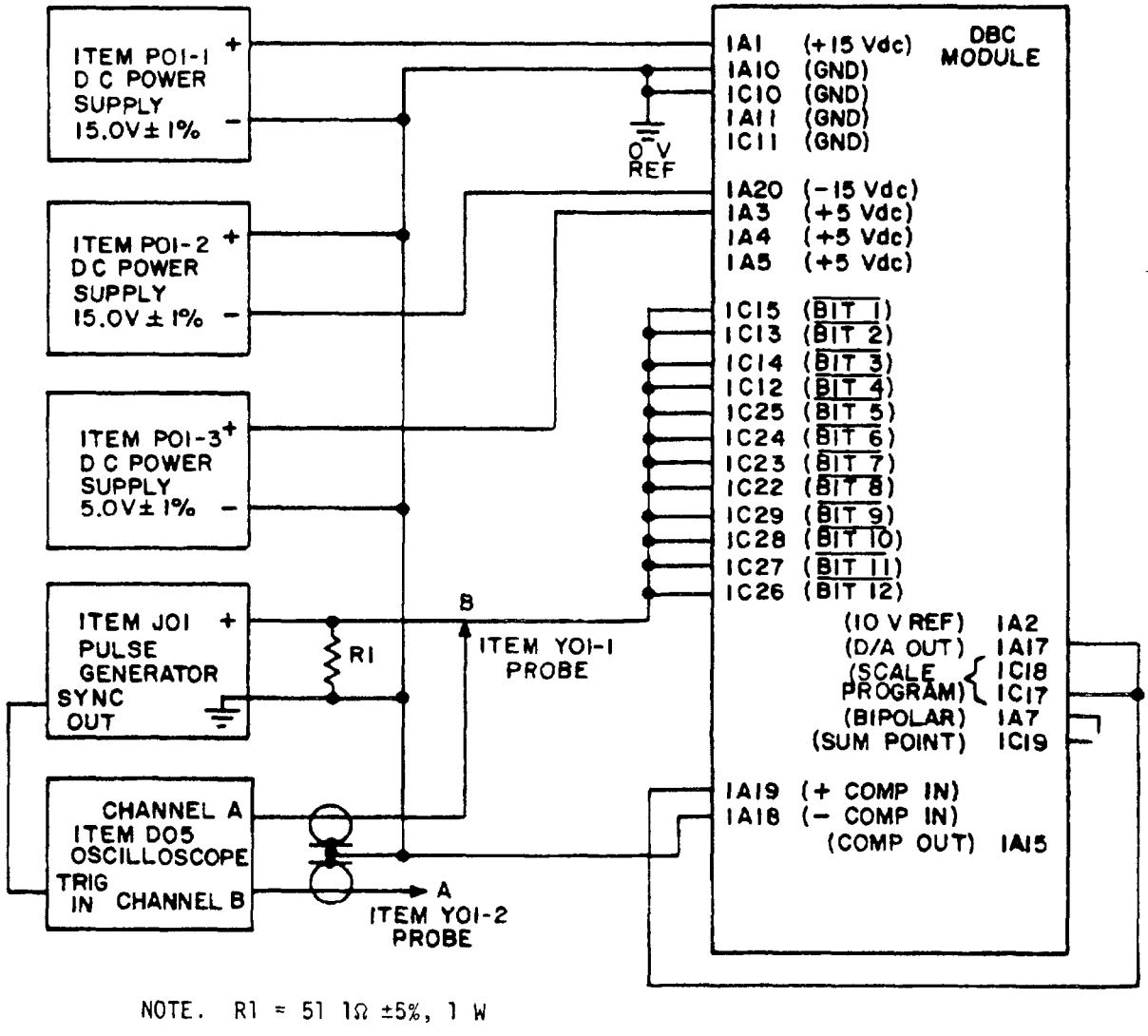
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NOTE Capacitance bridge (item C01) setup
Frequency = 1 kHz.
V_{pk} = 0.3 V maximum

*FIGURE 17 Power supply filter capacitance test circuit for table XIII.

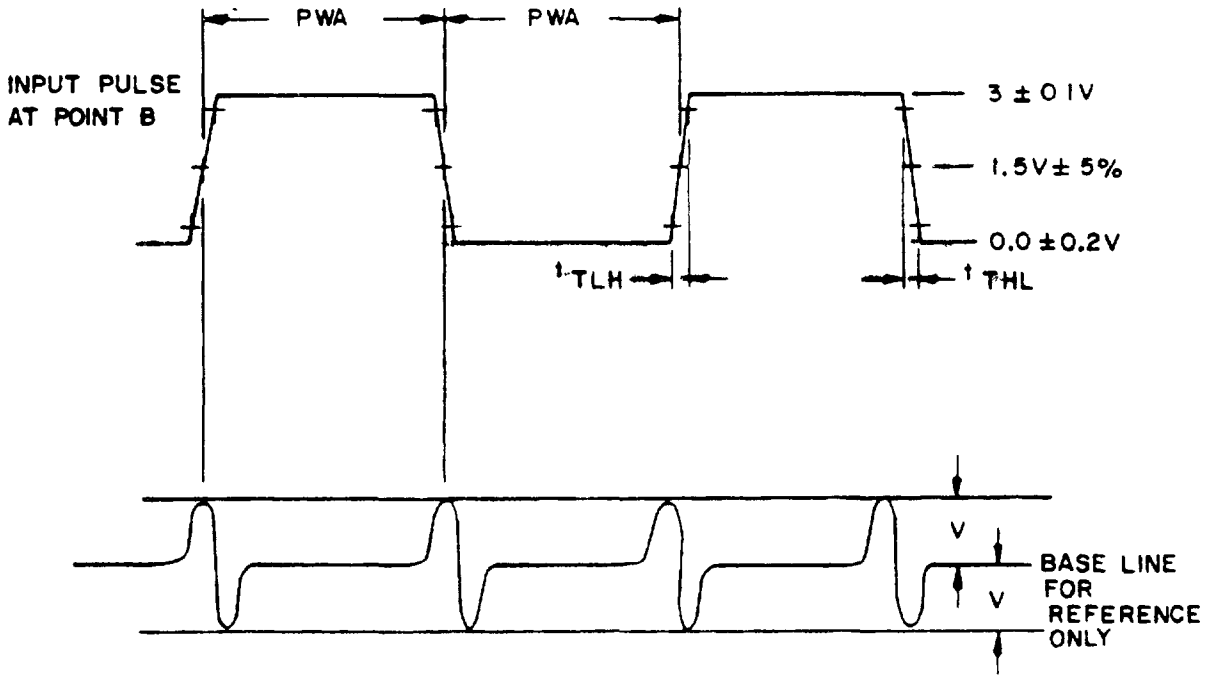
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NOTE. R1 = 51 1Ω ±5%, 1 W

FIGURE 18 Power supply transient amplitude test circuit for table XIV

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NOTES
1. Input pulse parameters
   PWA = 4 ±0.4 µs.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. The value V is measured from the baseline to the most negative excursion of the waveform for test conditions U1, U3, and U4.
3. The value V is measured from the baseline to the most positive excursion of the waveform for test condition U2

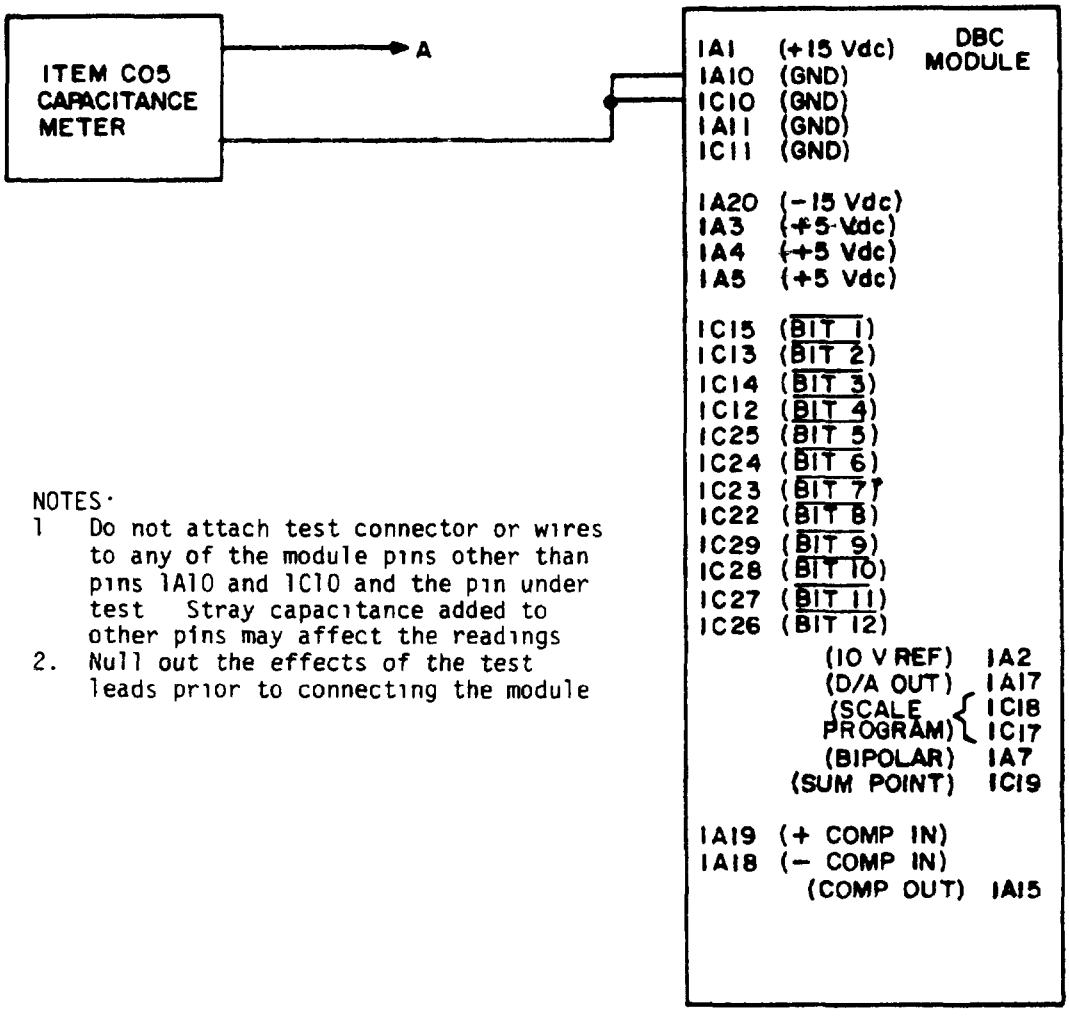
FIGURE 19. Power supply transient amplitude test circuit waveforms for table XIV

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TABLE XV Input capacitance test

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
V1	$\overline{\text{BIT1}}$ Input capacitance	1C15	2	15	pF	20
V2	$\overline{\text{BIT2}}$	1C13				
V3	$\overline{\text{BIT3}}$	1C14				
V4	$\overline{\text{BIT4}}$	1C12				
V5	$\overline{\text{BIT5}}$	1C25				
V6	$\overline{\text{BIT6}}$	1C24				
V7	$\overline{\text{BIT7}}$	1C23				
V8	$\overline{\text{BIT8}}$	1C22				
V9	$\overline{\text{BIT9}}$	1C29				
V10	$\overline{\text{BIT10}}$	1C28				
V11	$\overline{\text{BIT11}}$	1C27				
V12	$\overline{\text{BIT12}}$	1C26				
V13	- COMP IN	1A18				
V14	+ COMP IN	1A19				

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pins 1A10 and 1C10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 20. Input capacitance test circuit for table XV.

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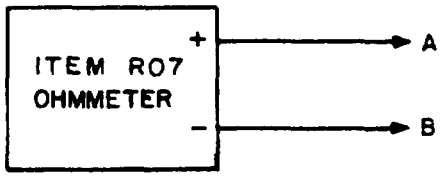
TABLE XVI Continuity test

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	Continuity (+5 V)	1A3	1A4	0	1	Ω	21
W2	Continuity (+5 V)	1A3	1A5				
W3	Continuity (GND)	1A10	1A11				
W4	Continuity (GND)	1A10	1C10				
W5	Continuity (GND)	1A10	1C11				
W6	Continuity (FRAME GND)	1C31	Frame				

TABLE XVII D/A output stability check test

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		Pin to pin	Min	Max		
X1	D/A output stability check, 0 - 10 V scale	1C18 to 1A17	No oscillation		-	22
X2	D/A output stability check, 0 - 5 V scale	1C18 to 1A17, 1C19 to 1C17				
X3	D/A output stability check, -10 to +10 V scale	1A7 to 1C19, 1C17 to 1A17				
X4	D/A output stability check, -5 to +5 V scale	1A7 to 1C19, 1C18 to 1A17				

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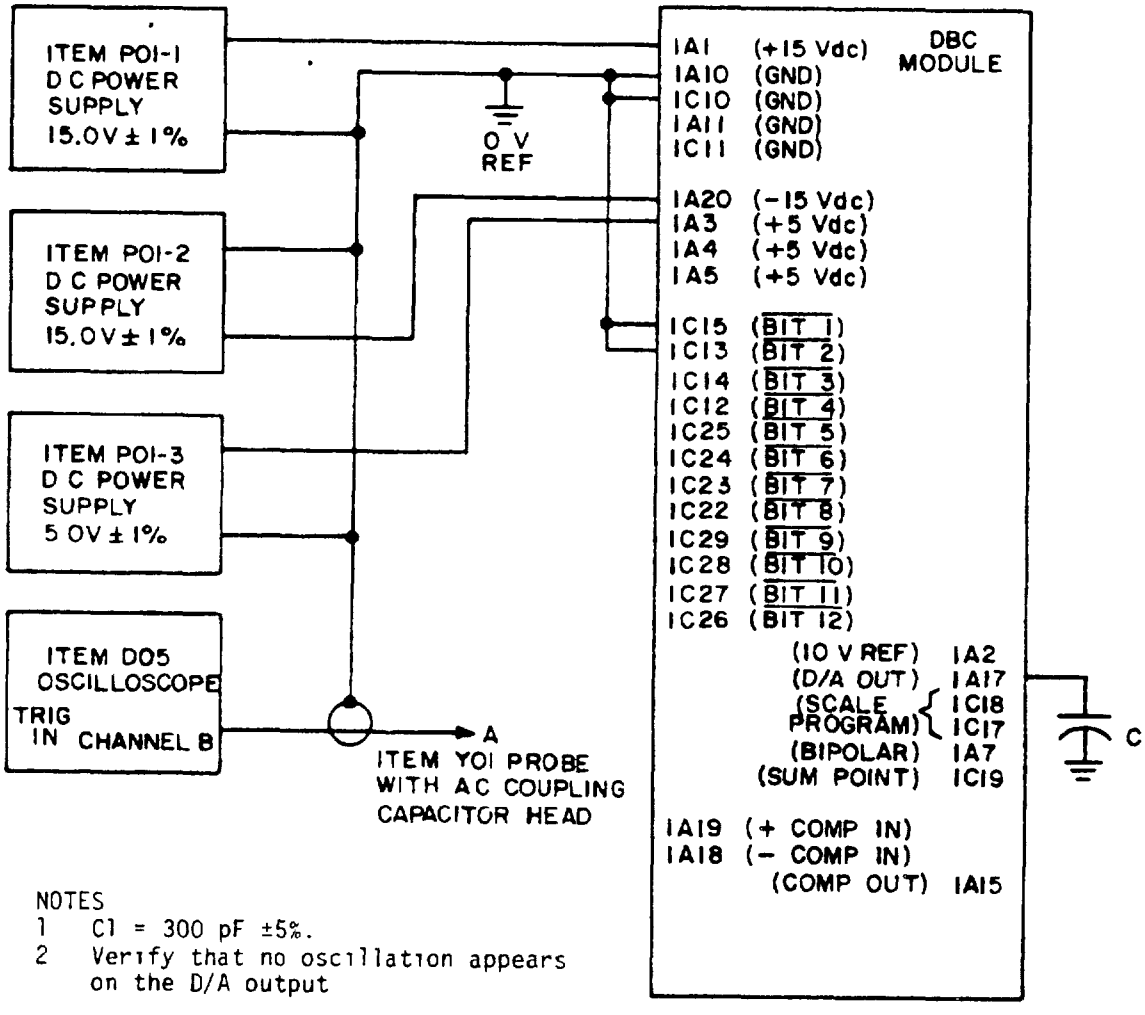
NOTE. For test condition W6 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal

DBC MODULE

1A1	(+15 Vdc)	
1A10	(GND)	
1C10	(GND)	
1A11	(GND)	
1C11	(GND)	
1A20	(-15 Vdc)	
1A3	(+5 Vdc)	
1A4	(+5 Vdc)	
1A5	(+5 Vdc)	
1C15	($\overline{\text{BIT 1}}$)	
1C13	($\overline{\text{BIT 2}}$)	
1C14	($\overline{\text{BIT 3}}$)	
1C12	($\overline{\text{BIT 4}}$)	
1C25	($\overline{\text{BIT 5}}$)	
1C24	($\overline{\text{BIT 6}}$)	
1C23	($\overline{\text{BIT 7}}$)	
1C22	($\overline{\text{BIT 8}}$)	
1C29	($\overline{\text{BIT 9}}$)	
1C28	($\overline{\text{BIT 10}}$)	
1C27	($\overline{\text{BIT 11}}$)	
1C26	(BIT 12)	
	(10 V REF)	1A2
	(D/A OUT)	1A17
	(SCALE PROGRAM)	1C18, 1C17
	(BIPOLAR)	1A7
	(SUM POINT)	1C19
1A19	(+ COMP IN)	
1A18	(- COMP IN)	
	(COMP OUT)	1A15

FIGURE 21. Continuity test circuit for table XVI.

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NOTES
1 C1 = 300 pF ±5%.
2 Verify that no oscillation appears on the D/A output

FIGURE 22 D/A output stability check test circuit for table XVII

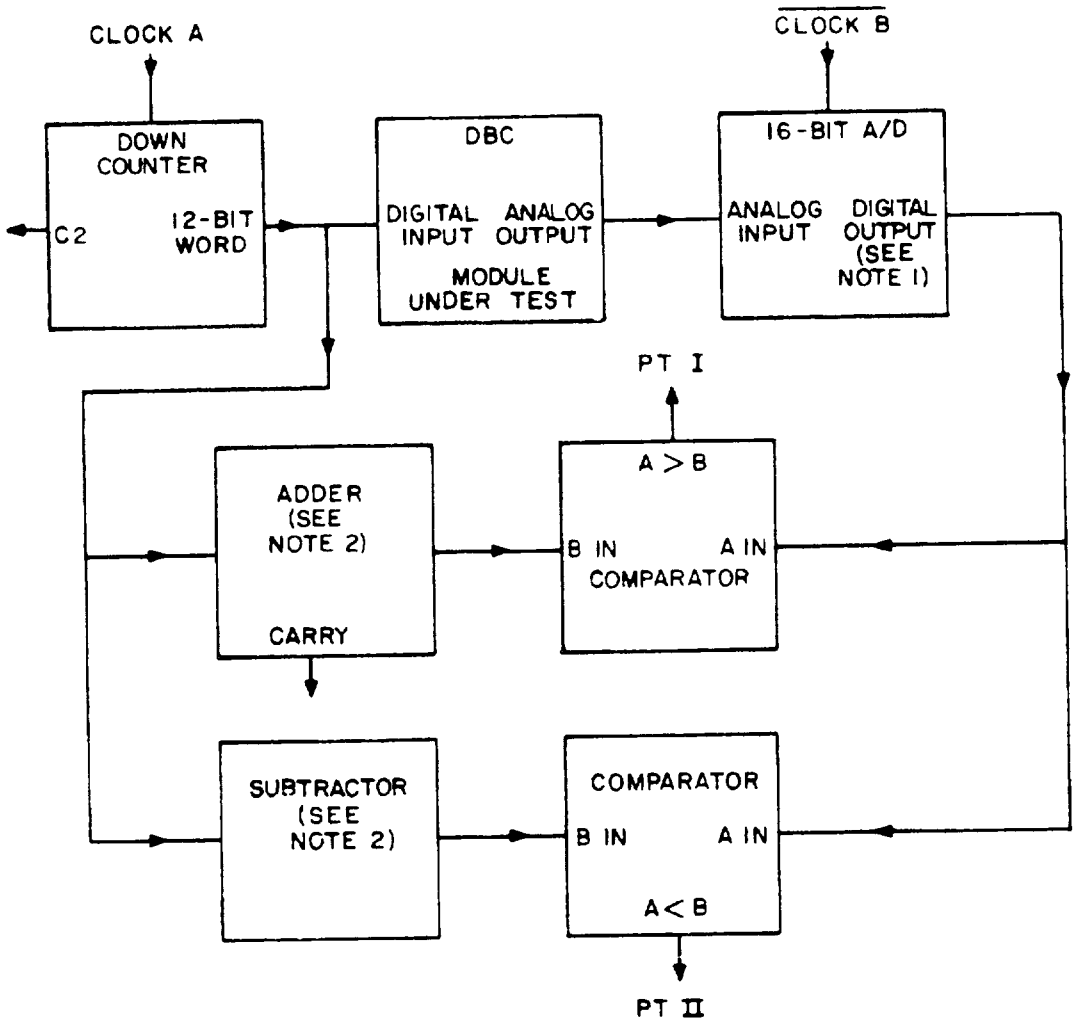
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* TABLE XVIII. Absolute accuracy, monotonicity, and linearity test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
Y 1/	Absolute accuracy	-2	+2	LSB	23,26,27
Z 1/	Monotonicity	-3/4	+1/2	LSB	24,26,27
AA 1/	Linearity	-1	+1	LSB	25,26,27

1/ Only one voltage scale is required for each test condition. Suggested scale is -10 to +10 V.

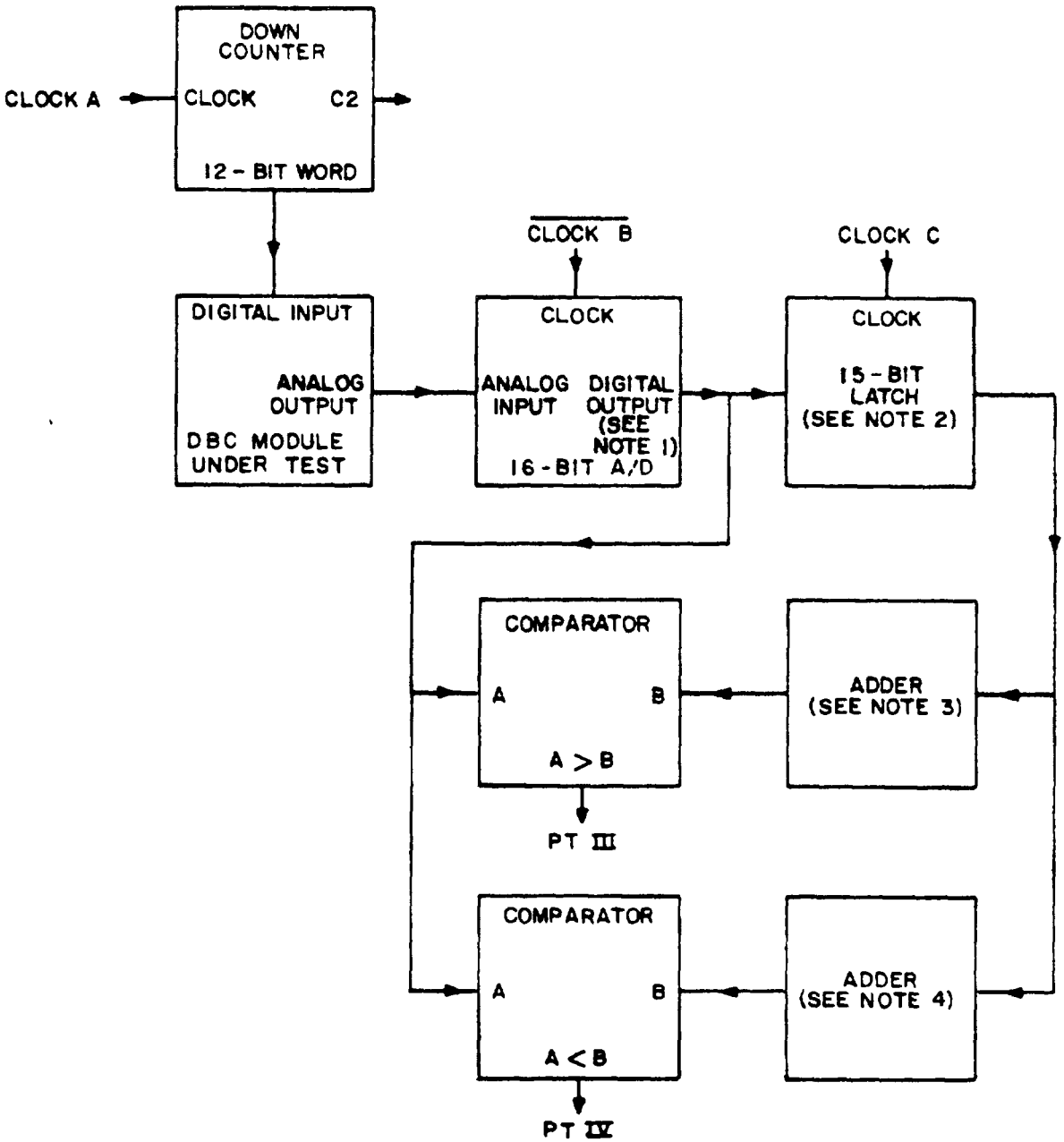
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NOTES:
1. The 15 most significant bits of the digital output are used.
2. Add and subtract 2 LSBs of the DBC input (12 bits).
3. C2, CARRY, PT I, and PT II go to figure 26.
4. CLOCK A and CLOCK B come from figure 26.

*FIGURE 23 Absolute accuracy test circuit for table XVIII

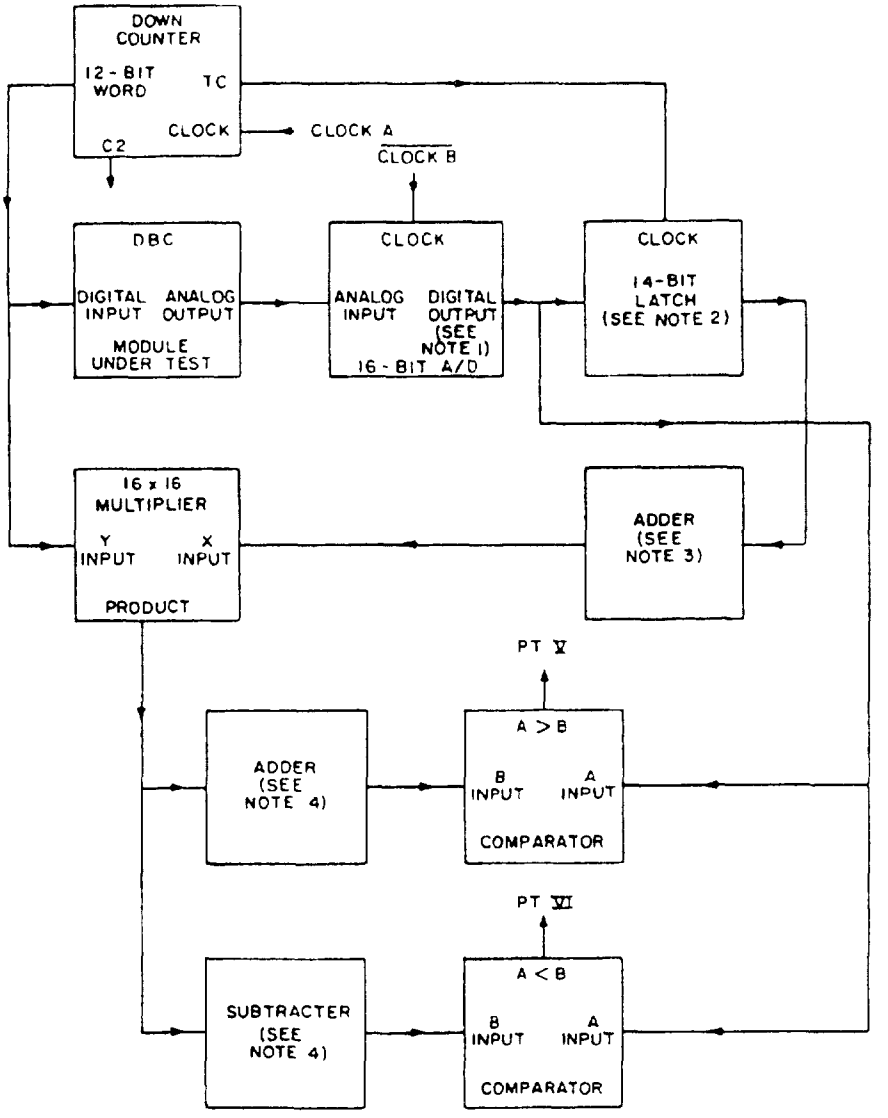
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NOTES.
1. The 15 most significant bits of the digital output are used
2 The latch is clocked at the same time as the counter. The previous word is therefore latched.
3. Bits corresponding to +1 -1/2 LSB of the DBC are added.
4 Bits corresponding to +1/4 LSB of the DBC are added.
5 PT III, PT IV, and C2 go to figure 26.
6. CLOCK A, $\overline{\text{CLOCK B}}$, and CLOCK C come from figure 26.

*FIGURE 24. Monotonicity test circuit for table XVIII.

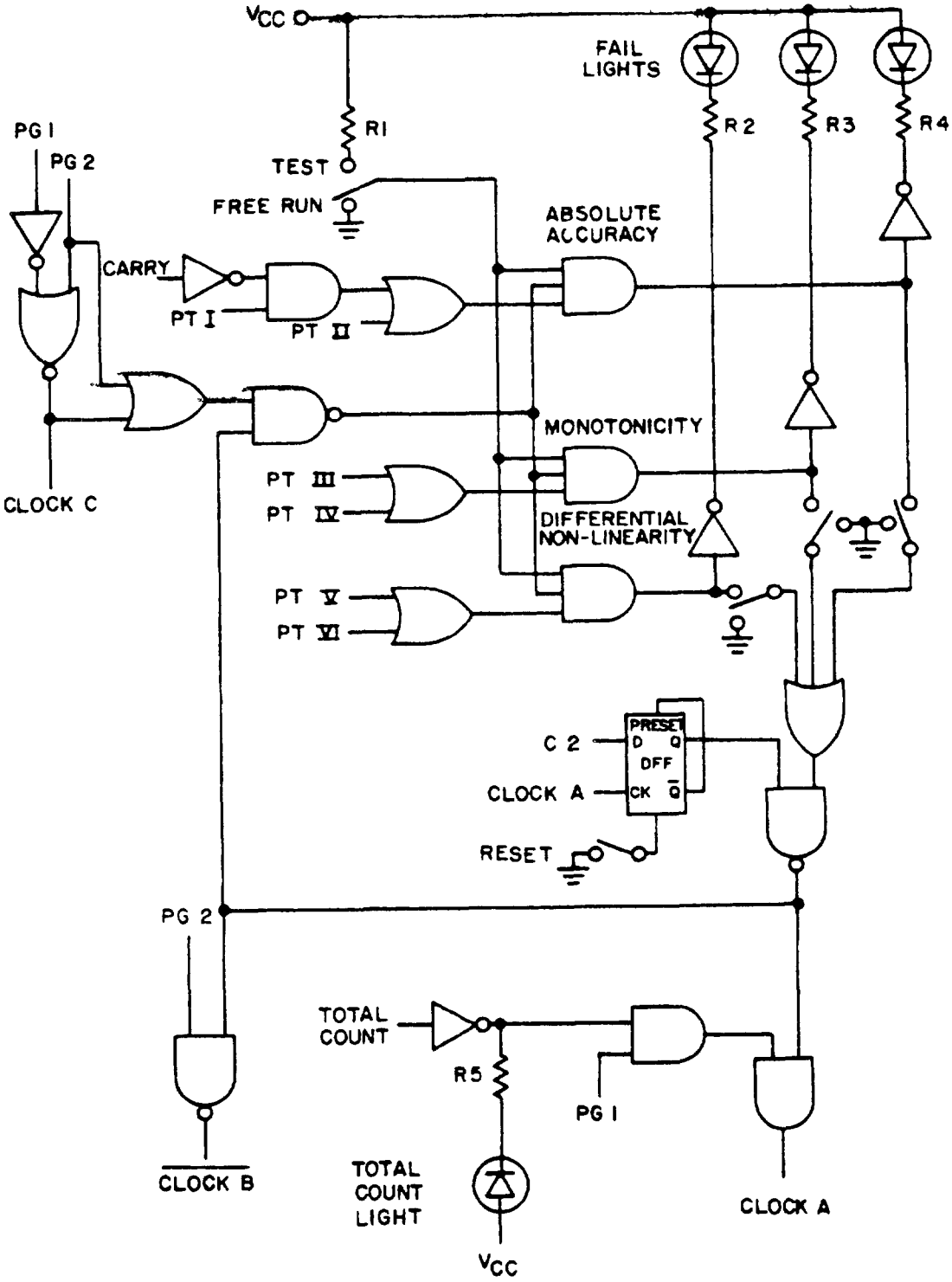
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NOTES:
1. The 14 most significant bits of the digital output are used.
2. The latch is clocked by the total count signal of the down counter. The last output of the A/D is latched. The counter must be cycled a siecond time to complete the test.
3. Bits are added to correspond to +2 LSB of the DBC.
4. Bits are added and subtracted to correspond to ±1 LBS of the DBC.
5. PT V, PT VI, and C2 go to figure 26.
6. CLOCK A and CLOCK B come from figure 26.

FIGURE 25. Linearity test circuit for table XVIII.

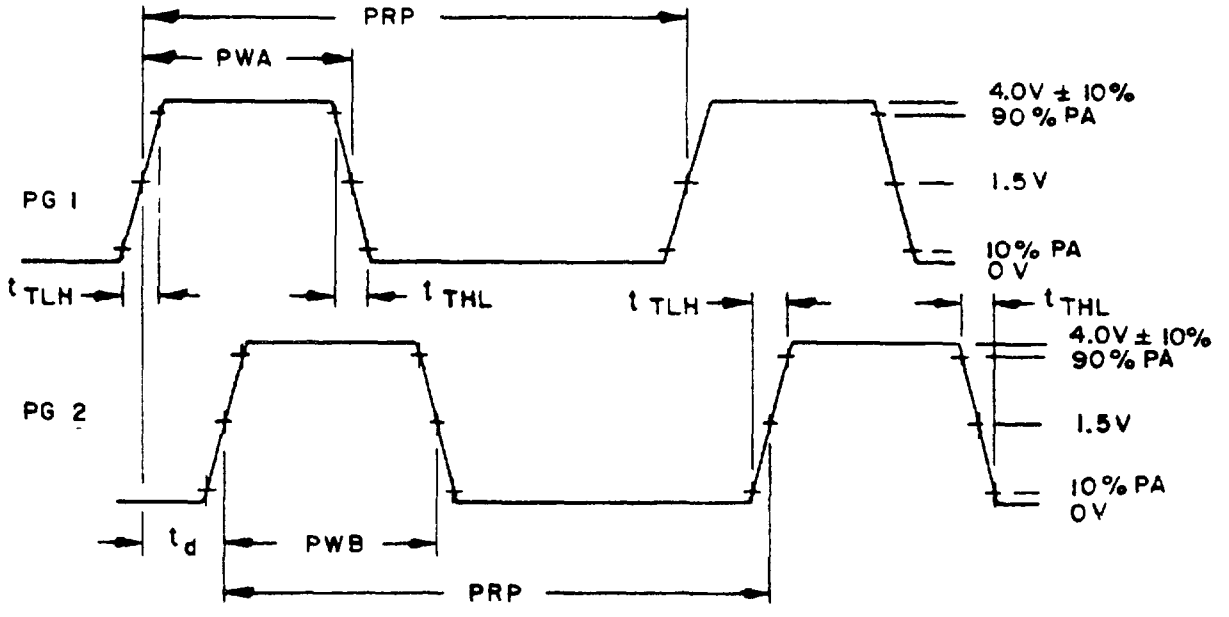
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NOTES
1. CLOCK A and $\overline{\text{CLOCK B}}$ go to figures 23 and 24. CLOCK C goes to figure 24.
2. PT I, PT II, and C2 come from figure 23.
3. PT III, PT IV, and C2 come from figure 24.
4. PT V, PT VI, and C2 come from figure 25.

* FIGURE 26. Absolute accuracy, monotonicity, and linearity test circuit for table XVIII.

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NOTES:
1 Input pulse parameters
PRP = 500 ±100 µs.
PWA = PWB = 200 ±40 µs.
t_d = 100 ±20 µs.
$t_{TLH} = t_{THL}$ = 50 ±10 ns
2 All voltages are referenced to 0 V

* FIGURE 27. Absolute accuracy, monotonicity, and linearity test circuit waveforms for table XVIII.

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